

Single_Cell_Pipeline_with_ScanPy

June 1, 2026

```
[ ]: !pip install scanpy
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Collecting scanpy
  Downloading scanpy-1.11.4-py3-none-any.whl.metadata (9.2 kB)
Collecting anndata>=0.8 (from scanpy)
  Downloading anndata-0.12.3-py3-none-any.whl.metadata (9.9 kB)
Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.15.0)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5.2)
Collecting legacy-api-wrap>=1.4.1 (from scanpy)
  Downloading legacy_api_wrap-1.4.1-py3-none-any.whl.metadata (2.1 kB)
Requirement already satisfied: matplotlib>=3.7.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy) (8.4.0)
Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.5)
Requirement already satisfied: numba>=0.57.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.60.0)
Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (2.0.2)
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from scanpy) (25.0)
Requirement already satisfied: pandas>=1.5.3 in /usr/local/lib/python3.12/dist-packages (from scanpy) (2.2.2)
Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.0.1)
Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.13)
Requirement already satisfied: scikit-learn>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.6.1)
Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.16.2)
Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.13.2)
Collecting session-info2 (from scanpy)
  Downloading session_info2-0.2.3-py3-none-any.whl.metadata (3.4 kB)
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Requirement already satisfied: statsmodels>=0.14.5 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from scanpy) (4.67.1)

Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)

Requirement already satisfied: umap-learn>=0.5.6 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.9.post2)

Collecting array-api-compat>=1.7.1 (from anndata>=0.8->scanpy)
 Downloading array_api_compat-1.12.0-py3-none-any.whl.metadata (2.5 kB)

Collecting zarr!=3.0.*,>=2.18.7 (from anndata>=0.8->scanpy)
 Downloading zarr-3.1.3-py3-none-any.whl.metadata (10 kB)

Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.3.3)

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)
(4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)
(2.9.0.post0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba>=0.57.1->scanpy) (0.43.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy)
(3.6.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.7.5->scanpy) (1.17.0)

Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
 Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)

Collecting numcodecs>=0.14 (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
 Downloading numcodecs-0.16.3-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.3 kB)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (6.0.3)

Collecting crc32c>=2.7 (from

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numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading crc32c-2.8-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.8
kB)
Downloading scanpy-1.11.4-py3-none-any.whl (2.1 MB)
      2.1/2.1 MB
27.8 MB/s eta 0:00:00
Downloading anndata-0.12.3-py3-none-any.whl (169 kB)
      169.3/169.3 kB
8.0 MB/s eta 0:00:00
Downloading legacy_api_wrap-1.4.1-py3-none-any.whl (10.0 kB)
Downloading session_info2-0.2.3-py3-none-any.whl (16 kB)
Downloading array_api_compat-1.12.0-py3-none-any.whl (58 kB)
      58.2/58.2 kB
2.8 MB/s eta 0:00:00
Downloading zarr-3.1.3-py3-none-any.whl (276 kB)
      276.4/276.4 kB
15.9 MB/s eta 0:00:00
Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)
Downloading
numcodecs-0.16.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (8.8
MB)
      8.8/8.8 MB
53.3 MB/s eta 0:00:00
Downloading crc32c-2.8-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (80 kB)
      80.0/80.0 kB
3.6 MB/s eta 0:00:00
Installing collected packages: session-info2, numcodecs, legacy-api-wrap,
donfig, crc32c, array-api-compat, zarr, anndata, scanpy
Successfully installed anndata-0.12.3 array-api-compat-1.12.0 crc32c-2.8
donfig-0.8.1.post1 legacy-api-wrap-1.4.1 numcodecs-0.16.3 scanpy-1.11.4 session-
info2-0.2.3 zarr-3.1.3

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```
[ ]: !pip install anndata
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Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
packages (0.12.3)
Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata) (1.12.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata) (3.15.0)
Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata) (1.4.1)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata) (8.4.0)
Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-
packages (from anndata) (2.0.2)

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Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata) (25.0)
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in
/usr/local/lib/python3.12/dist-packages (from anndata) (2.2.2)
Requirement already satisfied: scipy>=1.12 in /usr/local/lib/python3.12/dist-
packages (from anndata) (1.16.2)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from anndata) (3.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
packages (from zarr!=3.0.*,>=2.18.7->anndata) (0.8.1.post1)
Requirement already satisfied: numcodecs>=0.14 in
/usr/local/lib/python3.12/dist-packages (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata) (0.16.3)
Requirement already satisfied: typing-extensions>=4.9 in
/usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata)
(4.15.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata) (6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata) (2.8)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas!=2.1.2,>=2.1.0->anndata) (1.17.0)

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```
[ ]: !pip3 install igraph
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Collecting igraph
  Downloading
igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(4.4 kB)
Collecting texttable>=1.6.2 (from igraph)
  Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)
Downloading
igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.4 MB)
4.4/4.4 MB
70.1 MB/s eta 0:00:00
Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)
Installing collected packages: texttable, igraph
Successfully installed igraph-0.11.9 texttable-1.7.0

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```
[ ]: !pip install celltypist
```

Collecting celltypist

Downloading celltypist-1.7.1-py3-none-any.whl.metadata (45 kB)

0.0/45.1

kB ? eta -:--:--

45.1/45.1 kB 1.8

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Requirement already satisfied: numpy>=1.19.0 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (2.0.2)

Requirement already satisfied: pandas>=1.0.5 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.2.2)

Requirement already satisfied: scikit-learn>=0.24.1 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (1.6.1)

Requirement already satisfied: openpyxl>=3.0.4 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)

Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from celltypist) (8.3.0)

Requirement already satisfied: requests>=2.23.0 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (2.32.4)

Requirement already satisfied: scanpy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (1.11.4)

Collecting leidenalg>=0.9.0 (from celltypist)

Downloading leidenalg-0.10.2-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (10 kB)

Requirement already satisfied: igraph<0.12,>=0.10.0 in

/usr/local/lib/python3.12/dist-packages (from leidenalg>=0.9.0->celltypist) (0.11.9)

Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl>=3.0.4->celltypist) (2.0.0)

Requirement already satisfied: python-dateutil>=2.8.2 in

/usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist) (2025.2)

Requirement already satisfied: charset_normalizer<4,>=2 in

/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in

/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in

/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist) (2025.10.5)

Requirement already satisfied: anndata>=0.8 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.12.3)

Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (3.15.0)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (1.5.2)

Requirement already satisfied: legacy-api-wrap>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (1.4.1)

Requirement already satisfied: matplotlib>=3.7.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (3.10.0)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (8.4.0)

Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (3.5)

Requirement already satisfied: numba>=0.57.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.60.0)

Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (25.0)

Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (1.0.1)

Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.5.13)

Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (1.16.2)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.13.2)

Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.2.3)

Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.67.1)

Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.15.0)

Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.5.9.post2)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist) (3.6.0)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (3.1.3)

Requirement already satisfied: texttable>=1.6.2 in /usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->leidenalg>=0.9.0->celltypist) (1.7.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (3.2.5)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.57.1->scanpy>=1.7.0->celltypist) (0.43.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.16.3)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (6.0.3)

Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (2.8)

Downloading celltypist-1.7.1-py3-none-any.whl (7.3 MB)

7.3/7.3 MB

46.8 MB/s eta 0:00:00

Downloading
leidenalg-0.10.2-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.0 MB)

2.0/2.0 MB

43.0 MB/s eta 0:00:00

Installing collected packages: leidenalg, celltypist
Successfully installed celltypist-1.7.1 leidenalg-0.10.2

```
[ ]: !pip install decoupler
```

Collecting decoupler

Downloading decoupler-2.1.1-py3-none-any.whl.metadata (8.6 kB)

Collecting adjusttext (from decoupler)

Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)

Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-packages (from decoupler) (0.12.3)

Collecting docrep (from decoupler)

Downloading docrep-0.3.2.tar.gz (33 kB)

Preparing metadata (setup.py) ... done

Collecting marsilea (from decoupler)

Downloading marsilea-0.5.6-py3-none-any.whl.metadata (7.3 kB)

Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages (from decoupler) (0.60.0)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from decoupler) (2.32.4)

Collecting scipy<1.16 (from decoupler)

Downloading

scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)

62.0/62.0 kB

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Requirement already satisfied: session-info2 in

/usr/local/lib/python3.12/dist-packages (from decoupler) (0.2.3)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from decoupler) (4.67.1)

Requirement already satisfied: numpy<2.5,>=1.23.5 in

/usr/local/lib/python3.12/dist-packages (from scipy<1.16->decoupler) (2.0.2)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from adjusttext->decoupler) (3.10.0)

Requirement already satisfied: array-api-compat>=1.7.1 in

/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.12.0)

Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.15.0)

Requirement already satisfied: legacy-api-wrap in

/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.4.1)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (8.4.0)

Requirement already satisfied: packaging>=24.2 in

/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (25.0)

Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in

/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in

/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.1.3)

Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from docrep->decoupler) (1.17.0)

Collecting legendkit (from marsilea->decoupler)

 Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)

Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-packages (from marsilea->decoupler) (4.5.0)

Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (from marsilea->decoupler) (0.13.2)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2025.10.5)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.16.3)

Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from

```
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)
Requirement already satisfied: pyarrow>=10.0.1 in
/usr/local/lib/python3.12/dist-packages (from
pandas[parquet]->marsilea->decoupler) (18.1.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (2.8)
Downloading decoupler-2.1.1-py3-none-any.whl (117 kB)
```

118.0/118.0 kB

9.3 MB/s eta 0:00:00

Downloading

scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3 MB)

37.3/37.3 MB

46.9 MB/s eta 0:00:00

Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)

Downloading marsilea-0.5.6-py3-none-any.whl (84 kB)

85.0/85.0 kB

6.5 MB/s eta 0:00:00

Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)

Building wheels for collected packages: docrep

Building wheel for docrep (setup.py) ... done

Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876
sha256=d40f4d3e45d9520de1210f10a281b677e807126930b09ff6b308696d44f0a603

Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca
b57ce52da3e6fab7614836bd8c

Successfully built docrep

Installing collected packages: scipy, docrep, legendkit, adjustttext, marsilea,
decoupler

Attempting uninstall: scipy

Found existing installation: scipy 1.16.2

Uninstalling scipy-1.16.2:

Successfully uninstalled scipy-1.16.2

Successfully installed adjustttext-1.3.0 decoupler-2.1.1 docrep-0.3.2

legendkit-0.3.6 marsilea-0.5.6 scipy-1.15.3

```
[ ]: !pip install fa2-modified
```

Collecting fa2-modified

Downloading fa2_modified-0.3.10.tar.gz (474 kB)

0.0/474.3

kB ? eta -:--:--

471.0/474.3 kB

28.8 MB/s eta 0:00:01

474.3/474.3 kB

11.3 MB/s eta 0:00:00

```
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (1.15.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (4.67.1)
Building wheels for collected packages: fa2-modified
  Building wheel for fa2-modified (pyproject.toml) ... done
  Created wheel for fa2-modified:
    filename=fa2_modified-0.3.10-cp312-cp312-linux_x86_64.whl size=493274
    sha256=00f055e114ef390f908fe5959ae832343292017d406ba6b28e2ab4d186a73607
  Stored in directory: /root/.cache/pip/wheels/e7/b5/c2/e87afecc23dc78723045cde9
a733677ee2b3c1374a7ed3a5e8
Successfully built fa2-modified
Installing collected packages: fa2-modified
Successfully installed fa2-modified-0.3.10
```

```
[ ]: #Import core single cell datasets
```

```
import scanpy as sc
import anndata as ad
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/_utils/_init_.py:33:
FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
  from anndata import __version__ as anndata_version
/usr/local/lib/python3.12/dist-packages/scanpy/_init_.py:24: FutureWarning:
`__version__` is deprecated, use `importlib.metadata.version('anndata')`
instead.
  if Version(anndata.__version__) >= Version("0.11.0rc2"):
/usr/local/lib/python3.12/dist-packages/scanpy/readwrite.py:16: FutureWarning:
`__version__` is deprecated, use `importlib.metadata.version('anndata')`
instead.
  if Version(anndata.__version__) >= Version("0.11.0rc2"):
```

```
[ ]: !wget https://cf.10xgenomics.com/samples/cell-vdj/8.0.1/
↪10k_5p_Human_diseased_PBMC_ALL_Fresh/
↪10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.h5
```

```
--2025-10-20 01:35:33-- https://cf.10xgenomics.com/samples/cell-vdj/8.0.1/10k_5
p_Human_diseased_PBMC_ALL_Fresh/10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filte
red_feature_bc_matrix.h5
Resolving cf.10xgenomics.com (cf.10xgenomics.com)... 104.18.0.173, 104.18.1.173,
2606:4700::6812:1ad, ...
```

```

Connecting to cf.10xgenomics.com (cf.10xgenomics.com)|104.18.0.173|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 65402953 (62M) [application/x-hdf5]
Saving to:
'10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.h5'

10k_5p_Human_diseas 100%[=====>] 62.37M 17.0MB/s in 3.7s

2025-10-20 01:35:37 (17.0 MB/s) -
'10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.h5' saved
[65402953/65402953]

```

```

[ ]: fresh_blood_adata = sc.
      ↪read_10x_h5('10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.
      ↪h5')

```

```

/usr/local/lib/python3.12/dist-packages/anndata/_core/anndata.py:1794:
UserWarning: Variable names are not unique. To make them unique, call
`.var_names_make_unique`.
  utils.warn_names_duplicates("var")
/usr/local/lib/python3.12/dist-packages/anndata/_core/anndata.py:1794:
UserWarning: Variable names are not unique. To make them unique, call
`.var_names_make_unique`.
  utils.warn_names_duplicates("var")

```

```

[ ]: print(fresh_blood_adata)

```

```

AnnData object with n_obs × n_vars = 13853 × 38606
  var: 'gene_ids', 'feature_types', 'genome'

```

```

[ ]: fresh_blood_adata.shape

```

```

[ ]: (13853, 38606)

```

```

[ ]: #13853 cells
      #38606 genes

```

```

[ ]: fresh_blood_adata.var.head()

```

```

[ ]:
      gene_ids      feature_types  genome
DDX11L2  ENSG00000290825  Gene Expression  GRCh38
MIR1302-2HG  ENSG00000243485  Gene Expression  GRCh38
FAM138A  ENSG00000237613  Gene Expression  GRCh38
ENSG00000290826  ENSG00000290826  Gene Expression  GRCh38
OR4F5  ENSG00000186092  Gene Expression  GRCh38

```

```
[ ]: fresh_blood_adata.obs.head()
```

```
[ ]: Empty DataFrame
```

```
Columns: []
```

```
Index: [AAACCAAAGAACAGAC-1, AAACCAAAGCAAGATA-1, AAACCAAAGCTGGTTA-1,
AAACCAGCACCTAACG-1, AAACCAGCACGCGTTA-1]
```

```
[ ]: fresh_blood_adata.to_df()
```

```
[ ]:
```

	DDX11L2	MIR1302-2HG	FAM138A	ENSG00000290826	OR4F5	\
AAACCAAAGAACAGAC-1	0.0	0.0	0.0	0.0	0.0	
AAACCAAAGCAAGATA-1	0.0	0.0	0.0	0.0	0.0	
AAACCAAAGCTGGTTA-1	0.0	0.0	0.0	0.0	0.0	
AAACCAGCACCTAACG-1	0.0	0.0	0.0	0.0	0.0	
AAACCAGCACGCGTTA-1	0.0	0.0	0.0	0.0	0.0	
...	...	...	...	...	...	
GTTGTCTTCGTCCATT-2	0.0	0.0	0.0	0.0	0.0	
GTTGTGCAGAAGCCGA-2	0.0	0.0	0.0	0.0	0.0	
GTTGTGCAGATTACAC-2	0.0	0.0	0.0	0.0	0.0	
GTTGTGGGTAATGGGA-2	0.0	0.0	0.0	0.0	0.0	
GTTGTGGGTCGTCAGA-2	0.0	0.0	0.0	0.0	0.0	

	ENSG00000238009	ENSG00000239945	ENSG00000239906	\
AAACCAAAGAACAGAC-1	0.0	0.0	0.0	
AAACCAAAGCAAGATA-1	0.0	0.0	0.0	
AAACCAAAGCTGGTTA-1	0.0	0.0	0.0	
AAACCAGCACCTAACG-1	0.0	0.0	0.0	
AAACCAGCACGCGTTA-1	0.0	0.0	0.0	
...	...	...	...	
GTTGTCTTCGTCCATT-2	0.0	0.0	0.0	
GTTGTGCAGAAGCCGA-2	0.0	0.0	0.0	
GTTGTGCAGATTACAC-2	0.0	0.0	0.0	
GTTGTGGGTAATGGGA-2	0.0	0.0	0.0	
GTTGTGGGTCGTCAGA-2	0.0	0.0	0.0	

	ENSG00000241860	ENSG00000241599	...	ENSG00000275249	\
AAACCAAAGAACAGAC-1	0.0	0.0	...	0.0	
AAACCAAAGCAAGATA-1	0.0	0.0	...	0.0	
AAACCAAAGCTGGTTA-1	0.0	0.0	...	0.0	
AAACCAGCACCTAACG-1	0.0	0.0	...	0.0	
AAACCAGCACGCGTTA-1	0.0	0.0	...	0.0	
...	...	...	...	...	
GTTGTCTTCGTCCATT-2	0.0	0.0	...	0.0	
GTTGTGCAGAAGCCGA-2	0.0	0.0	...	0.0	
GTTGTGCAGATTACAC-2	0.0	0.0	...	0.0	
GTTGTGGGTAATGGGA-2	0.0	0.0	...	0.0	
GTTGTGGGTCGTCAGA-2	0.0	0.0	...	0.0	

	ENSG00000274792	ENSG00000274175	ENSG00000275869	\
AAACCAAAGAACAGAC-1	0.0	0.0	0.0	
AAACCAAAGCAAGATA-1	0.0	0.0	0.0	
AAACCAAAGCTGGTTA-1	0.0	0.0	0.0	
AAACCAGCACCTAACG-1	0.0	0.0	0.0	
AAACCAGCACGCGTTA-1	0.0	0.0	0.0	
...	...	...	...	
GTTGTCTTCGTCCATT-2	0.0	0.0	0.0	
GTTGTGCAGAAGCCGA-2	0.0	0.0	0.0	
GTTGTGCAGATTACAC-2	0.0	0.0	0.0	
GTTGTGGGTAATGGGA-2	0.0	0.0	0.0	
GTTGTGGGTCGTCAGA-2	0.0	0.0	0.0	
	ENSG00000273554	ENSG00000277836	ENSG00000278633	\
AAACCAAAGAACAGAC-1	0.0	0.0	0.0	
AAACCAAAGCAAGATA-1	0.0	0.0	0.0	
AAACCAAAGCTGGTTA-1	0.0	0.0	0.0	
AAACCAGCACCTAACG-1	0.0	0.0	0.0	
AAACCAGCACGCGTTA-1	0.0	0.0	0.0	
...	...	...	...	
GTTGTCTTCGTCCATT-2	0.0	0.0	0.0	
GTTGTGCAGAAGCCGA-2	0.0	0.0	0.0	
GTTGTGCAGATTACAC-2	0.0	0.0	0.0	
GTTGTGGGTAATGGGA-2	0.0	0.0	0.0	
GTTGTGGGTCGTCAGA-2	0.0	0.0	0.0	
	ENSG00000276017	ENSG00000278817	ENSG00000277196	
AAACCAAAGAACAGAC-1	0.0	0.0	0.0	
AAACCAAAGCAAGATA-1	0.0	0.0	0.0	
AAACCAAAGCTGGTTA-1	0.0	0.0	0.0	
AAACCAGCACCTAACG-1	0.0	0.0	0.0	
AAACCAGCACGCGTTA-1	0.0	0.0	0.0	
...	...	...	...	
GTTGTCTTCGTCCATT-2	0.0	0.0	0.0	
GTTGTGCAGAAGCCGA-2	0.0	0.0	0.0	
GTTGTGCAGATTACAC-2	0.0	0.0	0.0	
GTTGTGGGTAATGGGA-2	0.0	0.0	0.0	
GTTGTGGGTCGTCAGA-2	0.0	0.0	0.0	

[13853 rows x 38606 columns]

```
[ ]: fresh_blood_adata.obs_keys()
```

/tmp/ipython-input-923068421.py:1: FutureWarning: Use obs (e.g. `k` in `adata.obs`` or ``str(adata.obs.columns.tolist())``) instead of `AnnData.obs_keys`, `AnnData.obs_keys` is deprecated and will be removed in the future.

```
fresh_blood_adata.obs_keys()
```

```
[ ]: []
```

```
[ ]: fresh_blood_adata.uns_keys()
```

```
/tmp/ipython-input-119558665.py:1: FutureWarning: Use uns (e.g. `k in adata.uns`  
or `sorted(adata.uns)` instead of AnnData.uns_keys, AnnData.uns_keys is  
deprecated and will be removed in the future.
```

```
fresh_blood_adata.uns_keys()
```

```
[ ]: []
```

```
[ ]: # A useful step for older datasets  
fresh_blood_adata.var_names_make_unique()  
fresh_blood_adata.obs_names_make_unique()
```

```
[ ]: fresh_blood_adata.var
```

```
[ ]:
      gene_ids      feature_types  genome
DDX11L2      ENSG00000290825  Gene Expression  GRCh38
MIR1302-2HG    ENSG00000243485  Gene Expression  GRCh38
FAM138A        ENSG00000237613  Gene Expression  GRCh38
ENSG00000290826  ENSG00000290826  Gene Expression  GRCh38
OR4F5          ENSG00000186092  Gene Expression  GRCh38
...           ...           ...           ...
ENSG00000277836  ENSG00000277836  Gene Expression  GRCh38
ENSG00000278633  ENSG00000278633  Gene Expression  GRCh38
ENSG00000276017  ENSG00000276017  Gene Expression  GRCh38
ENSG00000278817  ENSG00000278817  Gene Expression  GRCh38
ENSG00000277196  ENSG00000277196  Gene Expression  GRCh38
```

```
[38606 rows x 3 columns]
```

```
[ ]: fresh_blood_adata.obs
```

```
[ ]: Empty DataFrame
```

```
Columns: []
```

```
Index: [AAACCAAAGAACAGAC-1, AAACCAAAGCAAGATA-1, AAACCAAAGCTGGTTA-1,  
AAACCAGCACCTAACG-1, AAACCAGCACGCGTTA-1, AAACCAGCACTAGGAA-1, AAACCATTCATTCTA-1,  
AAACCATTCATTGCT-1, AAACCATTCGGTGGTT-1, AAACCCATCAAACTAC-1, AAACCCATCACGACAC-1,  
AAACCCATCCAATCAG-1, AAACCCATCCGGTCAG-1, AAACCGACAATTCGTA-1, AAACCGACAGGGTTCC-1,  
AAACCGCTCAGTTTGG-1, AAACCGCTCCCACTAT-1, AAACCGGAGAGCGTAG-1, AAACCGGAGGTAATAG-1,  
AAACCTCCACGGTGTC-1, AAACCTCCAGTATAGG-1, AAACCTGTCAAGCAAA-1, AAACCTGTCCCTTCTC-1,  
AAACGATCAAAGTACG-1, AAACGATCACTCGGTA-1, AAACGATCAGCATGGG-1, AAACGCACAAGTGC-1,  
AAACGCGAGCCCACAG-1, AAACGCTGTACGCATG-1, AAACGCTGTCAGTAGT-1, AAACGAGTCAGCTCT-1,  
AAACGGAGTGCTATGG-1, AAACGGCCACCGTAAT-1, AAACGGTAGCACCACG-1, AAACGTAAGGCAACAG-1,  
AAACTAACAATTACGC-1, AAATACTCGGCGTAT-1, AAATACTCGTCAATG-1, AAATAAGAGAAAGTGC-1,
```

```
AAACTCCCAACAAGAT-1, AAACCTCCCAATAGGGC-1, AAACCTCCCAATCTTAG-1, AAACCTCCACCTAGGC-1,
AAACTCCCAGTTGAAG-1, AAACCTGCGTACTGCGA-1, AAACCTGCGTCTTAGGG-1, AAACCTGCGTGTCTAGG-1,
AAACTTATCGCCTGCT-1, AAAGCCACAACCTAAC-1, AAAGCCACACGTCGTA-1, AAAGCCACAGCTAGAA-1,
AAAGCCGAGAATACGA-1, AAAGCCGAGCTACAGC-1, AAAGCCTGTAACAGGA-1, AAAGCCTGTATTGGTA-1,
AAAGCCTGTGCGGTTA-1, AAAGCCTGTGCTTCCG-1, AAAGCCTGTGGCTTAG-1, AAAGCCTGTGGTTTCC-1,
AAAGCGAGTCCTCATA-1, AAAGCGAGTCGTGCCC-1, AAAGCTAAGCTGACAC-1, AAAGCTCGTCATTTCCG-1,
AAAGGAACACTCCACT-1, AAAGGAGAGCCCTTAT-1, AAAGGAGAGGGTCTGT-1, AAAGGAGAGGTAGAGT-1,
AAAGGAGAGGTATTGT-1, AAAGGCCCAATAATTC-1, AAAGGCCACGTACAT-1, AAAGGGGCAGGCATAA-1,
AAAGGGTTCAATTCCC-1, AAAGGTATCGTACATG-1, AAAGGTCAGAACGTGG-1, AAAGGTCAGACCTGTA-1,
AAAGGTCAGCTCGAGT-1, AAAGTCAAGGGATTAA-1, AAAGTGACAGAGGGATT-1, AAAGTTACAACAGCGG-1,
AAAGTTACACGCGTTA-1, AAAGTTACAGTATGCT-1, AAAGTTCTCCGTCATT-1, AAAGTTCTCGGAAACG-1,
AAATCAACACATCGGC-1, AAATCAACACGAACGG-1, AAATCAACAGCTCTAG-1, AAATCACTCCAGCTCC-1,
AAATCAGAGAGTCAGG-1, AAATCCAGTCCGTGTT-1, AAATCCAGTCTTCTAG-1, AAATCCGTCAGTCAGC-1,
AAATCCGTCGCTAACAA-1, AAATCCGTCGCGTCAG-1, AAATCTATCAAAGGGT-1, AAATCTATCGTTGTCT-1,
AAATCTGGTAAGCATC-1, AAATCTGGTATTACCT-1, AAATCTGGTCCAAAGT-1, AAATCTGGTGTAAAGTC-1,
AAATCTTCAAGAGGGA-1, ...]
```

```
[13853 rows x 0 columns]
```

```
[ ]: fresh_blood_adata.var_names
```

```
[ ]: Index(['DDX11L2', 'MIR1302-2HG', 'FAM138A', 'ENSG00000290826', 'OR4F5',
          'ENSG00000238009', 'ENSG00000239945', 'ENSG00000239906',
          'ENSG00000241860', 'ENSG00000241599',
          ...,
          'ENSG00000275249', 'ENSG00000274792', 'ENSG00000274175',
          'ENSG00000275869', 'ENSG00000273554', 'ENSG00000277836',
          'ENSG00000278633', 'ENSG00000276017', 'ENSG00000278817',
          'ENSG00000277196'],
          dtype='object', length=38606)
```

```
[ ]: fresh_blood_adata.var['MT'] = fresh_blood_adata.var_names.str.startswith("MT-")
fresh_blood_adata.var['RIBO'] = fresh_blood_adata.var_names.str.
    ↪startswith("RPS", "RPL")
fresh_blood_adata.var['HB'] = fresh_blood_adata.var_names.str.
    ↪startswith("^HB[^(P)]")
```

```
[ ]: fresh_blood_adata.var['MT']
```

```
[ ]: DDX11L2          False
MIR1302-2HG         False
FAM138A             False
ENSG00000290826     False
OR4F5               False
...
ENSG00000277836     False
ENSG00000278633     False
```



```

ENSG00000276017    False
ENSG00000278817    False
ENSG00000277196    False
Name: MT, Length: 38606, dtype: bool

```

```

[ ]: mt_genes = fresh_blood_adata.var[fresh_blood_adata.var['MT']]
     mt_genes

```

```

[ ]:
      gene_ids    feature_types    genome    MT    RIBO    HB
MT-ND1  ENSG00000198888  Gene Expression  GRCh38  True  False  False
MT-ND2  ENSG00000198763  Gene Expression  GRCh38  True  False  False
MT-CO1  ENSG00000198804  Gene Expression  GRCh38  True  False  False
MT-CO2  ENSG00000198712  Gene Expression  GRCh38  True  False  False
MT-ATP8  ENSG00000228253  Gene Expression  GRCh38  True  False  False
MT-ATP6  ENSG00000198899  Gene Expression  GRCh38  True  False  False
MT-CO3  ENSG00000198938  Gene Expression  GRCh38  True  False  False
MT-ND3  ENSG00000198840  Gene Expression  GRCh38  True  False  False
MT-ND4L  ENSG00000212907  Gene Expression  GRCh38  True  False  False
MT-ND4  ENSG00000198886  Gene Expression  GRCh38  True  False  False
MT-ND5  ENSG00000198786  Gene Expression  GRCh38  True  False  False
MT-ND6  ENSG00000198695  Gene Expression  GRCh38  True  False  False
MT-CYB  ENSG00000198727  Gene Expression  GRCh38  True  False  False

```

```

[ ]: sc.pp.calculate_qc_metrics(
      fresh_blood_adata, qc_vars=["MT", 'RIBO', 'HB'], inplace=True, log1p=True
    )

```

```

[ ]: fresh_blood_adata.obs.head()

```

```

[ ]:
      n_genes_by_counts  log1p_n_genes_by_counts  total_counts  \
AAACCAAAGAACAGAC-1      3351          8.117312      9437.0
AAACCAAAGCAAGATA-1      2718          7.908019      9637.0
AAACCAAAGCTGGTTA-1      3262          8.090402     10236.0
AAACCAGCACCTAACG-1      3184          8.066208      9737.0
AAACCAGCACGCGTTA-1      3342          8.114624     11524.0

      log1p_total_counts  pct_counts_in_top_50_genes  \
AAACCAAAGAACAGAC-1      9.152499          22.719084
AAACCAAAGCAAGATA-1      9.173469          35.872159
AAACCAAAGCTGGTTA-1      9.233764          29.054318
AAACCAGCACCTAACG-1      9.183791          26.887132
AAACCAGCACGCGTTA-1      9.352274          28.089205

      pct_counts_in_top_100_genes  pct_counts_in_top_200_genes  \
AAACCAAAGAACAGAC-1      30.772491          41.178341
AAACCAAAGCAAGATA-1      49.175054          58.285774
AAACCAAAGCTGGTTA-1      39.576006          49.267292

```

AAACCAGCACCTAACG-1	36.664270	47.047345
AAACCAGCACGCGTTA-1	40.758417	51.292954

	pct_counts_in_top_500_genes	total_counts_MT \
AAACCAAAGAACAGAC-1	56.575183	420.0
AAACCAAAGCAAGATA-1	69.959531	351.0
AAACCAAAGCTGGTTA-1	62.289957	472.0
AAACCAGCACCTAACG-1	60.778474	349.0
AAACCAGCACGCGTTA-1	64.213815	501.0

	log1p_total_counts_MT	pct_counts_MT	total_counts_RIBO \
AAACCAAAGAACAGAC-1	6.042633	4.450567	479.0
AAACCAAAGCAAGATA-1	5.863631	3.642212	1455.0
AAACCAAAGCTGGTTA-1	6.159095	4.611176	1048.0
AAACCAGCACCTAACG-1	5.857933	3.584266	696.0
AAACCAGCACGCGTTA-1	6.218600	4.347449	1302.0

	log1p_total_counts_RIBO	pct_counts_RIBO	total_counts_HB \
AAACCAAAGAACAGAC-1	6.173786	5.075766	0.0
AAACCAAAGCAAGATA-1	7.283448	15.098059	0.0
AAACCAAAGCTGGTTA-1	6.955593	10.238374	0.0
AAACCAGCACCTAACG-1	6.546785	7.147993	0.0
AAACCAGCACGCGTTA-1	7.172425	11.298161	0.0

	log1p_total_counts_HB	pct_counts_HB
AAACCAAAGAACAGAC-1	0.0	0.0
AAACCAAAGCAAGATA-1	0.0	0.0
AAACCAAAGCTGGTTA-1	0.0	0.0
AAACCAGCACCTAACG-1	0.0	0.0
AAACCAGCACGCGTTA-1	0.0	0.0

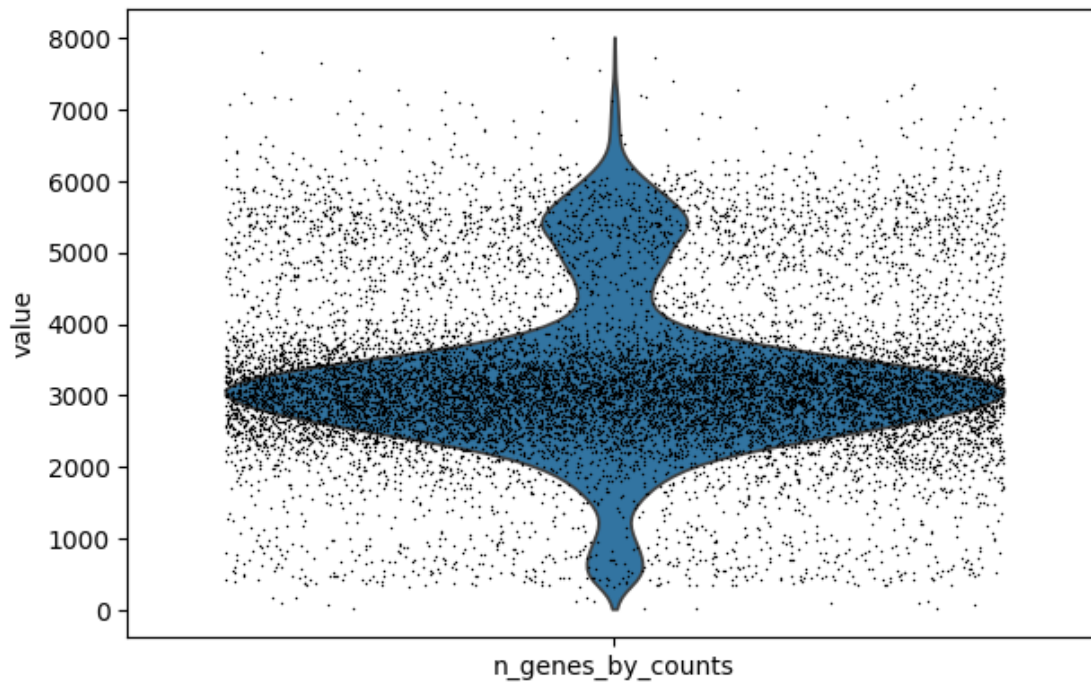
```
[ ]: fresh_blood_adata.var.head()
```

```
[ ]:
      gene_ids  feature_types  genome  MT  RIBO \
DDX11L2  ENSG00000290825  Gene Expression  GRCh38  False  False
MIR1302-2HG  ENSG00000243485  Gene Expression  GRCh38  False  False
FAM138A  ENSG00000237613  Gene Expression  GRCh38  False  False
ENSG00000290826  ENSG00000290826  Gene Expression  GRCh38  False  False
OR4F5  ENSG00000186092  Gene Expression  GRCh38  False  False
```

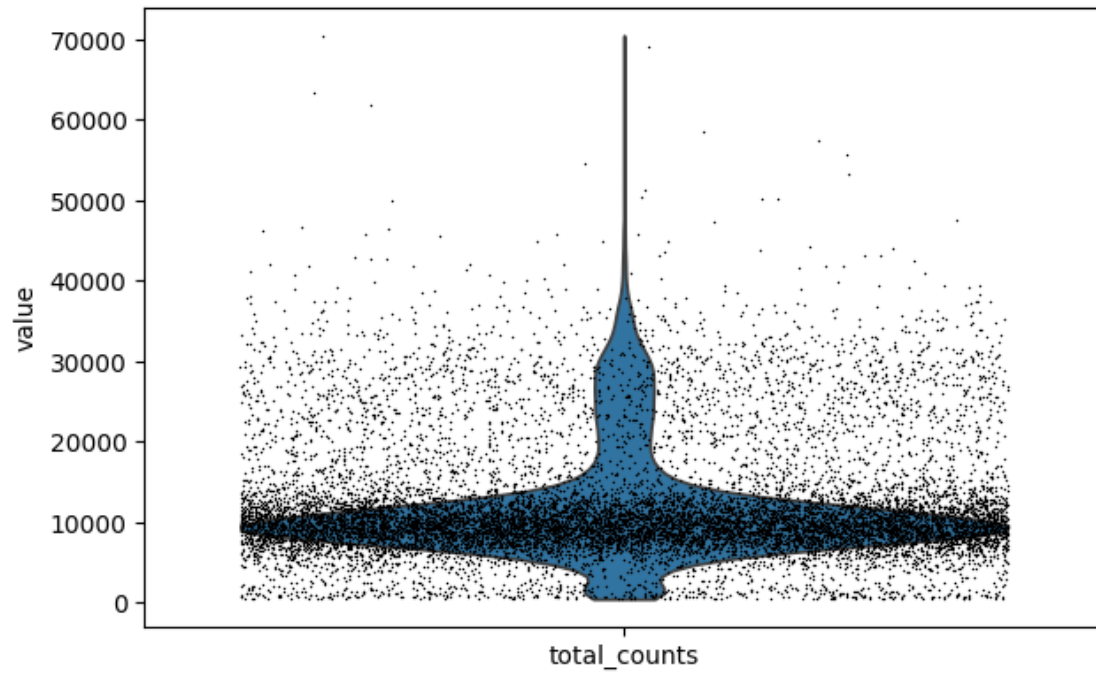
	HB	n_cells_by_counts	mean_counts	log1p_mean_counts \
DDX11L2	False	7	0.000505	0.000505
MIR1302-2HG	False	1	0.000072	0.000072
FAM138A	False	0	0.000000	0.000000
ENSG00000290826	False	0	0.000000	0.000000
OR4F5	False	0	0.000000	0.000000

	pct_dropout_by_counts	total_counts	log1p_total_counts
DDX11L2	99.949469	7.0	2.079442
MIR1302-2HG	99.992781	1.0	0.693147
FAM138A	100.000000	0.0	0.000000
ENSG00000290826	100.000000	0.0	0.000000
OR4F5	100.000000	0.0	0.000000

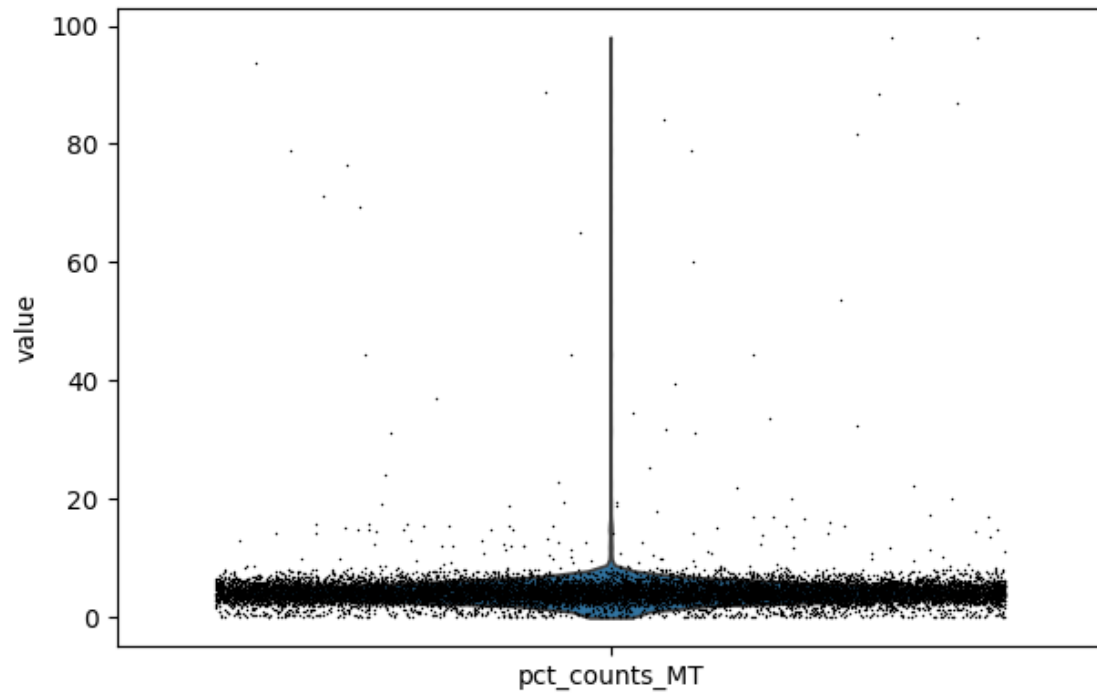
```
[ ]: sc.pl.violin(
    fresh_blood_adata,
    ["n_genes_by_counts"],
    jitter=0.4,
    multi_panel=False,
)
```



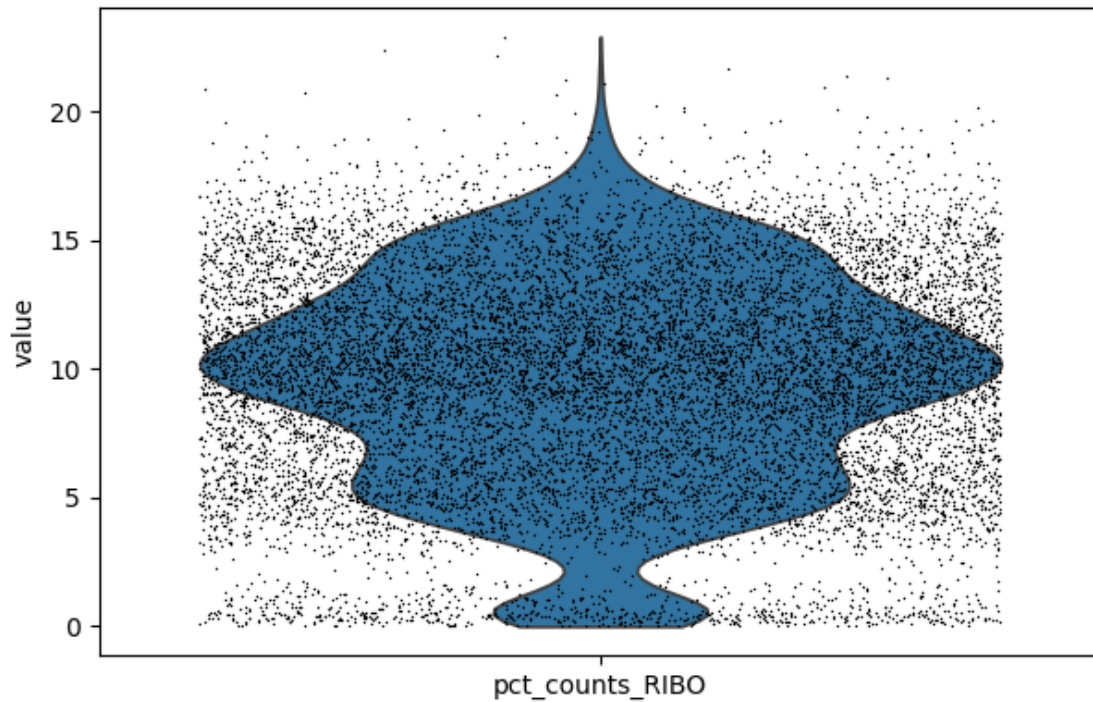
```
[ ]: sc.pl.violin(
    fresh_blood_adata,
    ["total_counts"],
    jitter=0.4,
    multi_panel=False,
)
```



```
[ ]: sc.pl.violin(  
    fresh_blood_adata,  
    ["pct_counts_MT"],  
    jitter=0.4,  
    multi_panel=False,  
)
```

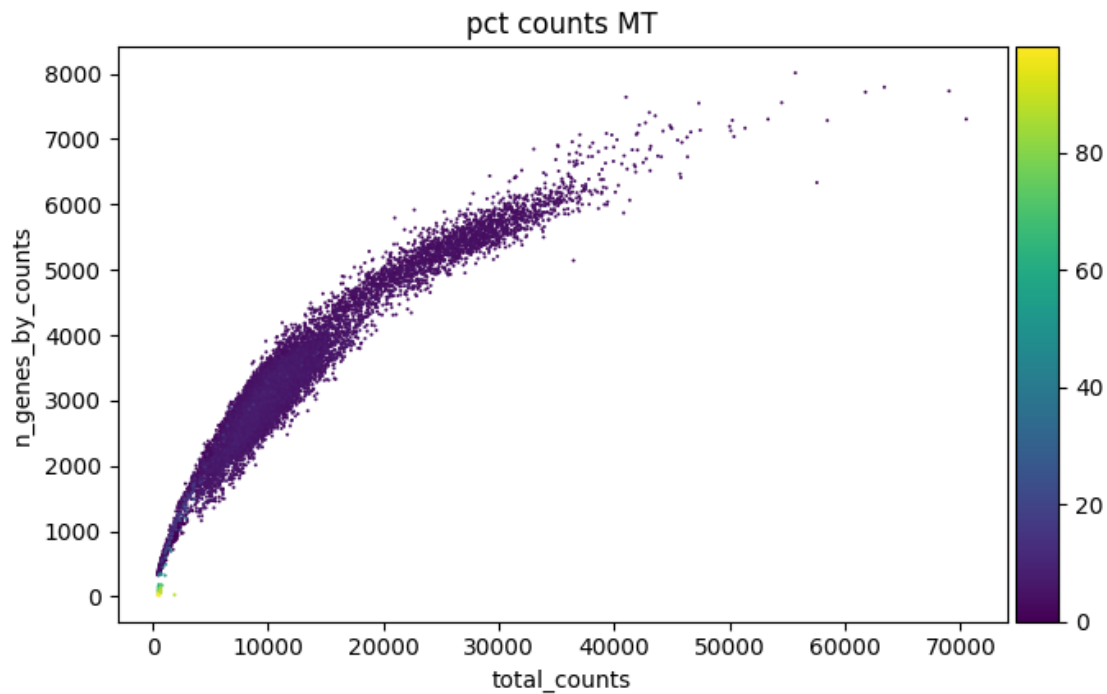


```
[ ]: sc.pl.violin(  
    fresh_blood_adata,  
    ["pct_counts_RIBO"],  
    jitter=0.4,  
    multi_panel=False,  
)
```



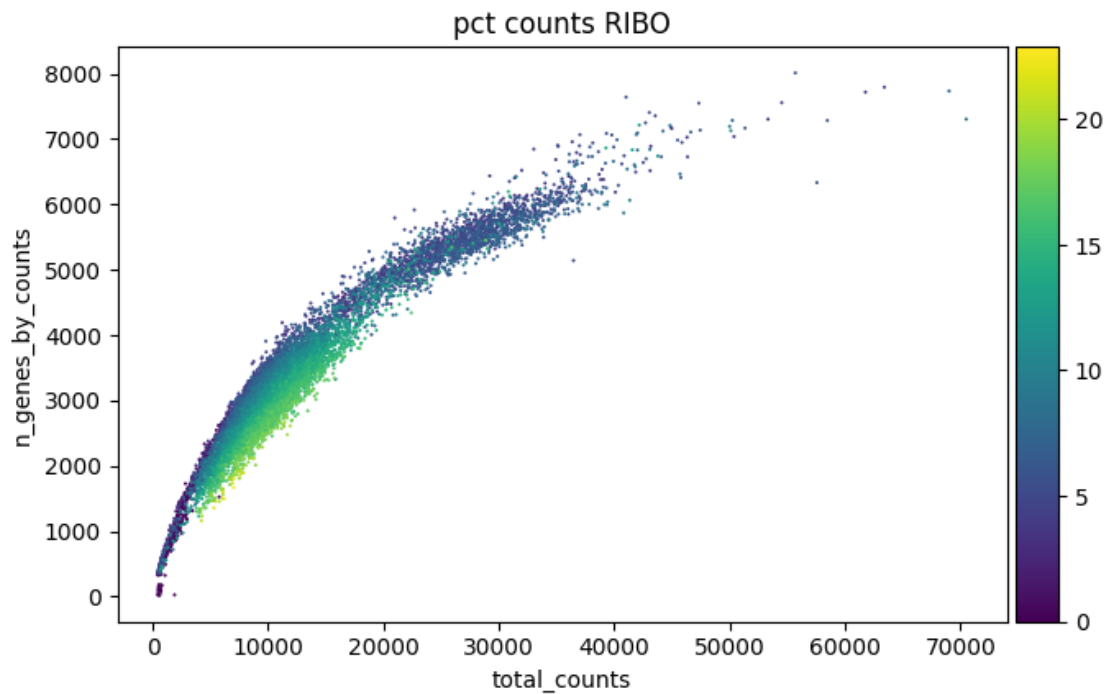
```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
↳ color="pct_counts_MT")
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/plotting/_anndata.py:397:
FutureWarning: Use obs (e.g. `k in adata.obs` or
`str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys, AnnData.obs_keys
is deprecated and will be removed in the future.
    if key in adata.obs_keys():
```



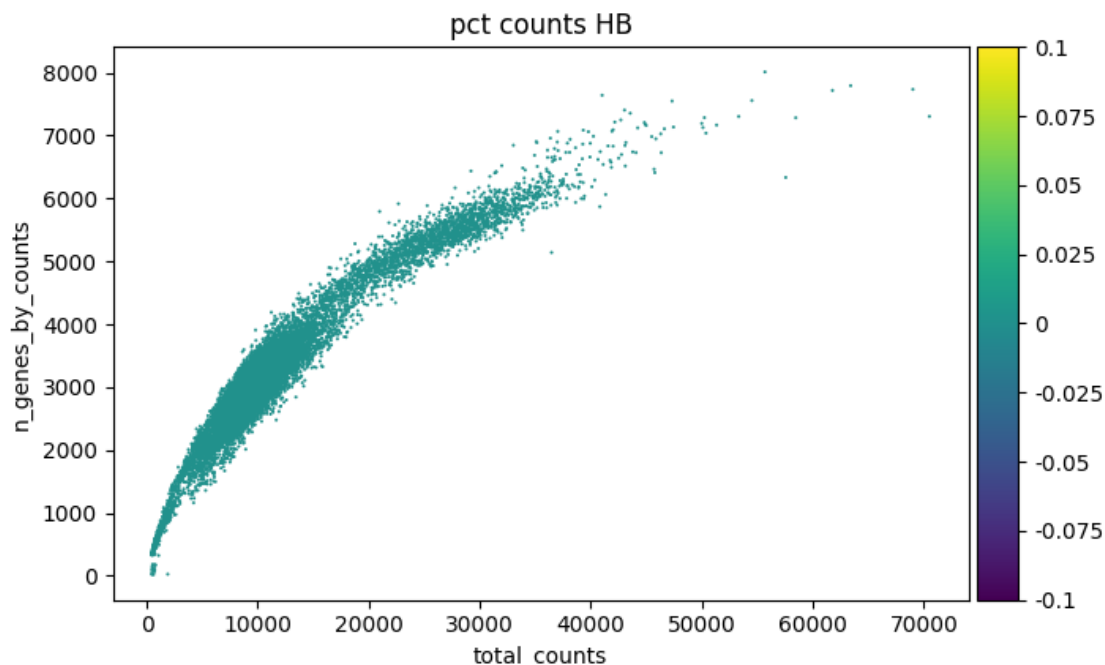
```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
    ↪color="pct_counts_RIB0")
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/plotting/_anndata.py:397:
FutureWarning: Use obs (e.g. `k in adata.obs` or
`str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys, AnnData.obs_keys
is deprecated and will be removed in the future.
    if key in adata.obs_keys():
```



```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
    ↪color="pct_counts_HB")
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/plotting/_anndata.py:397:
FutureWarning: Use obs (e.g. `k in adata.obs` or
`str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys, AnnData.obs_keys
is deprecated and will be removed in the future.
    if key in adata.obs_keys():
```

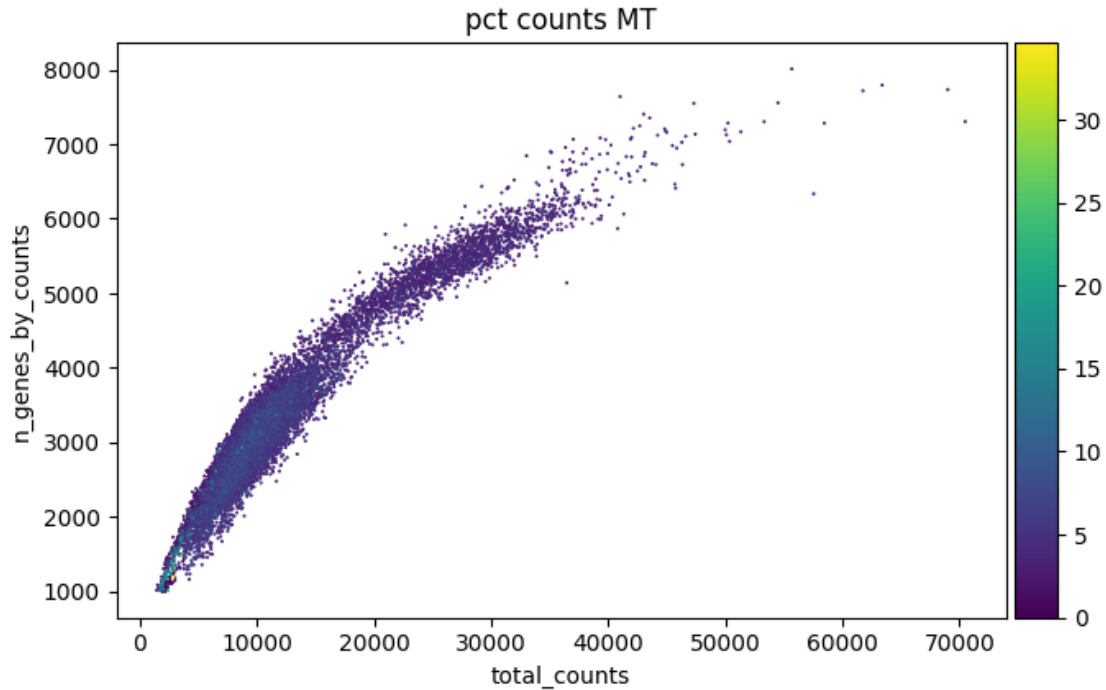
```
[ ]: """
    Additionally, it is important to note that for datasets with multiple batches,
    quality control should be performed for each sample individually as quality
    control thresholds can vary substantially between batches.
    """
```

```
[ ]: '\nAdditionally, it is important to note that for datasets with multiple
batches,\nquality control should be performed for each sample individually as
quality\ncontrol thresholds can vary substantially between batches.\n'
```

```
[ ]: sc.pp.filter_cells(fresh_blood_adata, min_genes=1000)
sc.pp.filter_genes(fresh_blood_adata, min_cells=1000)
```

```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
    ↪color="pct_counts_MT")
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/plotting/_anndata.py:397:
FutureWarning: Use obs (e.g. `k in adata.obs` or
`str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys, AnnData.obs_keys
is deprecated and will be removed in the future.
    if key in adata.obs_keys():
```



```
[ ]: #doublet detection
##Identifying doublets is crucial as they can lead to misclassifications or
##distortions in downstream analysis steps
```

```
[ ]: sc.pp.scrublet(fresh_blood_adata) #if you have multiple batch samples, you can
    ↪ specify it with batch_key="sample"
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/neighbors/_init_.py:430:
FutureWarning: Use obsm (e.g. `k in adata.obsm` or `adata.obsm.keys() | {'u'}`)
instead of AnnData.obsm_keys, AnnData.obsm_keys is deprecated and will be
removed in the future.
    if "X_diffmap" in adata.obsm_keys():
```

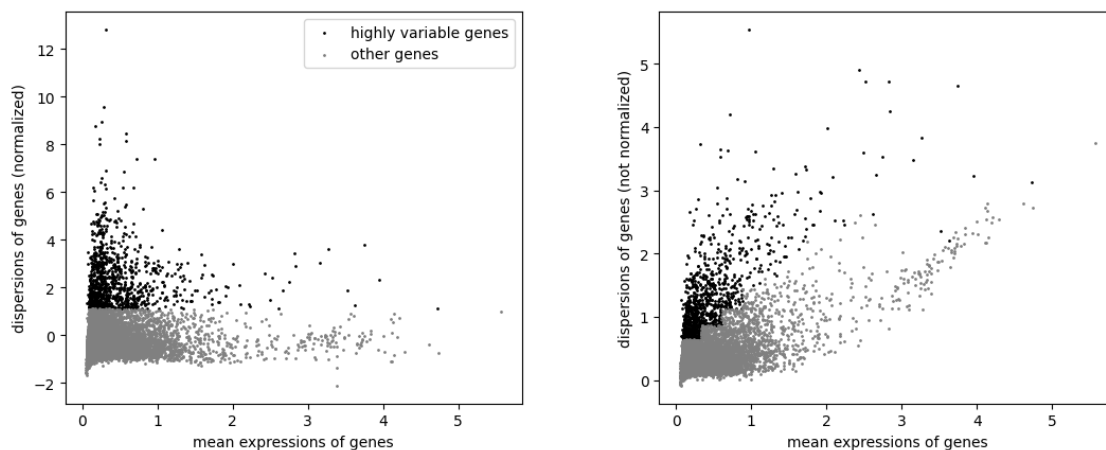
```
[ ]: #Normalization
```

```
[ ]: # Save a copy of the data
fresh_blood_adata.layers["counts"] = fresh_blood_adata.X.copy()
```

```
[ ]: # Normalizing to median total counts
sc.pp.normalize_total(fresh_blood_adata)
# Logarithmize the data
sc.pp.log1p(fresh_blood_adata)
```

```
[ ]: #Feature selection
#selecting the top 1000 most variable genes
sc.pp.highly_variable_genes(fresh_blood_adata, n_top_genes=1000)
```

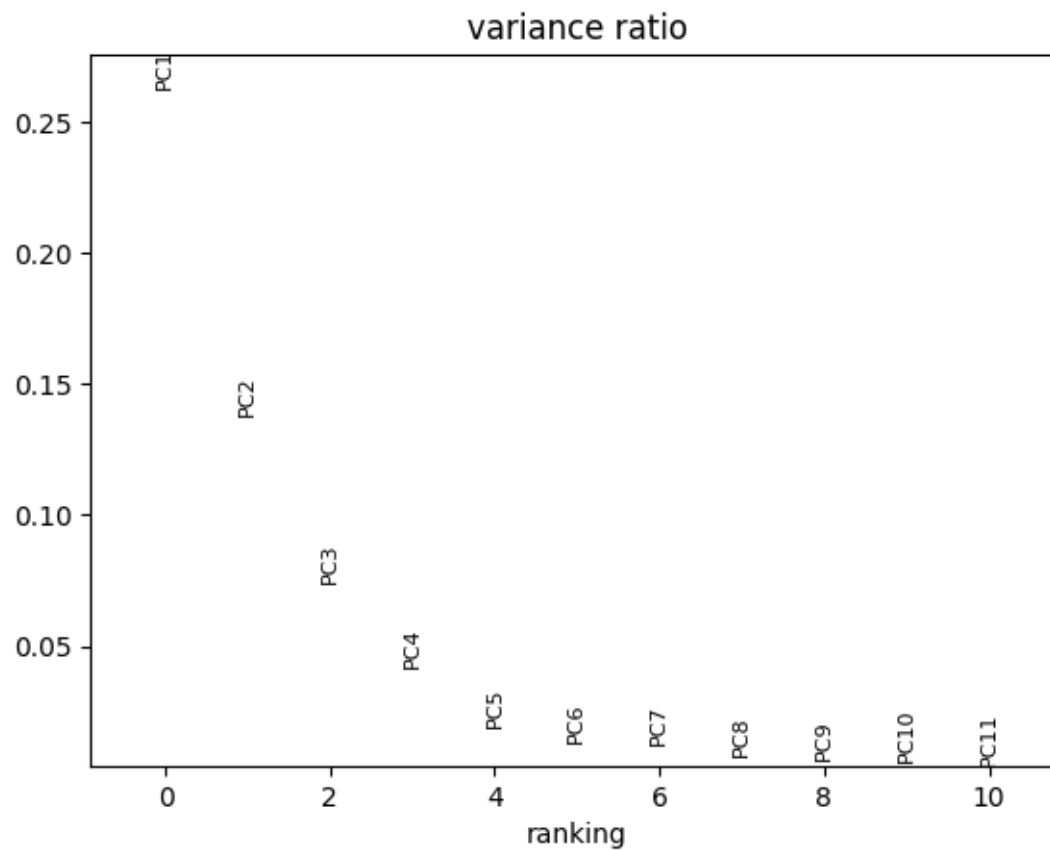
```
[ ]: sc.pl.highly_variable_genes(fresh_blood_adata )
#left is normalized
#right is not
```



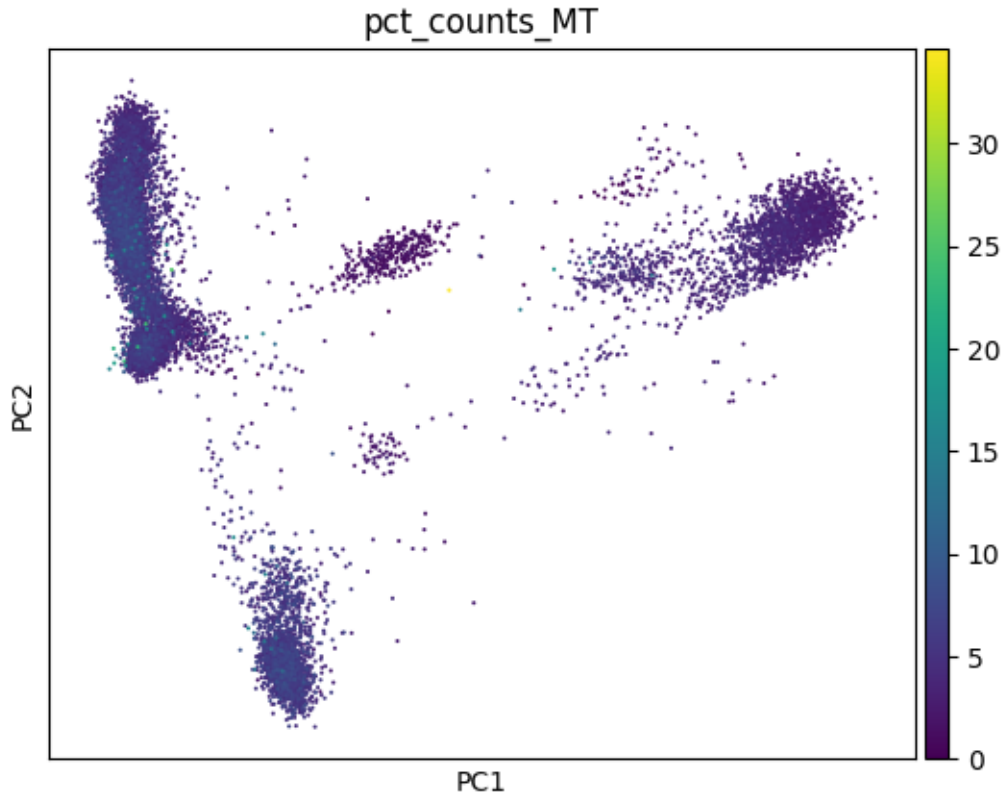
```
[ ]: sc.tl.pca(fresh_blood_adata)
```

```
/usr/local/lib/python3.12/dist-
packages/scanpy/preprocessing/_pca/__init__.py:245: FutureWarning: `__version__`
is deprecated, use `importlib.metadata.version('anndata')` instead.
  Version(ad.__version__) < Version("0.9")
```

```
[ ]: sc.pl.pca_variance_ratio(fresh_blood_adata, n_pcs=10, log=False)
```



```
[ ]: sc.pl.pca(  
    fresh_blood_adata,  
    color=["pct_counts_MT"]  
)
```



```
[ ]: sc.pp.neighbors(fresh_blood_adata, n_neighbors=10, n_pcs=10)
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/neighbors/_init_.py:430:
FutureWarning: Use obsm (e.g. `k` in adata.obsm` or `adata.obsm.keys() | {'u'}`)
instead of AnnData.obsm_keys, AnnData.obsm_keys is deprecated and will be
removed in the future.
    if "X_diffmap" in adata.obsm_keys():
```

```
[ ]: fresh_blood_adata.obs.head()
```

```
[ ]:
```

	n_genes_by_counts	log1p_n_genes_by_counts	total_counts	\
AAACCAAAGAACAGAC-1	3351	8.117312	9437.0	
AAACCAAAGCAAGATA-1	2718	7.908019	9637.0	
AAACCAAAGCTGGTTA-1	3262	8.090402	10236.0	
AAACCAGCACCTAACG-1	3184	8.066208	9737.0	
AAACCAGCACGCGTTA-1	3342	8.114624	11524.0	

	log1p_total_counts	pct_counts_in_top_50_genes	\
AAACCAAAGAACAGAC-1	9.152499	22.719084	
AAACCAAAGCAAGATA-1	9.173469	35.872159	
AAACCAAAGCTGGTTA-1	9.233764	29.054318	

AAACCAGCACCTAACG-1	9.183791	26.887132
AAACCAGCACGCGTTA-1	9.352274	28.089205

	pct_counts_in_top_100_genes	pct_counts_in_top_200_genes \
AAACCAAAGAACAGAC-1	30.772491	41.178341
AAACCAAAGCAAGATA-1	49.175054	58.285774
AAACCAAAGCTGGTTA-1	39.576006	49.267292
AAACCAGCACCTAACG-1	36.664270	47.047345
AAACCAGCACGCGTTA-1	40.758417	51.292954

	pct_counts_in_top_500_genes	total_counts_MT \
AAACCAAAGAACAGAC-1	56.575183	420.0
AAACCAAAGCAAGATA-1	69.959531	351.0
AAACCAAAGCTGGTTA-1	62.289957	472.0
AAACCAGCACCTAACG-1	60.778474	349.0
AAACCAGCACGCGTTA-1	64.213815	501.0

	loglp_total_counts_MT	pct_counts_MT	total_counts_RIBO \
AAACCAAAGAACAGAC-1	6.042633	4.450567	479.0
AAACCAAAGCAAGATA-1	5.863631	3.642212	1455.0
AAACCAAAGCTGGTTA-1	6.159095	4.611176	1048.0
AAACCAGCACCTAACG-1	5.857933	3.584266	696.0
AAACCAGCACGCGTTA-1	6.218600	4.347449	1302.0

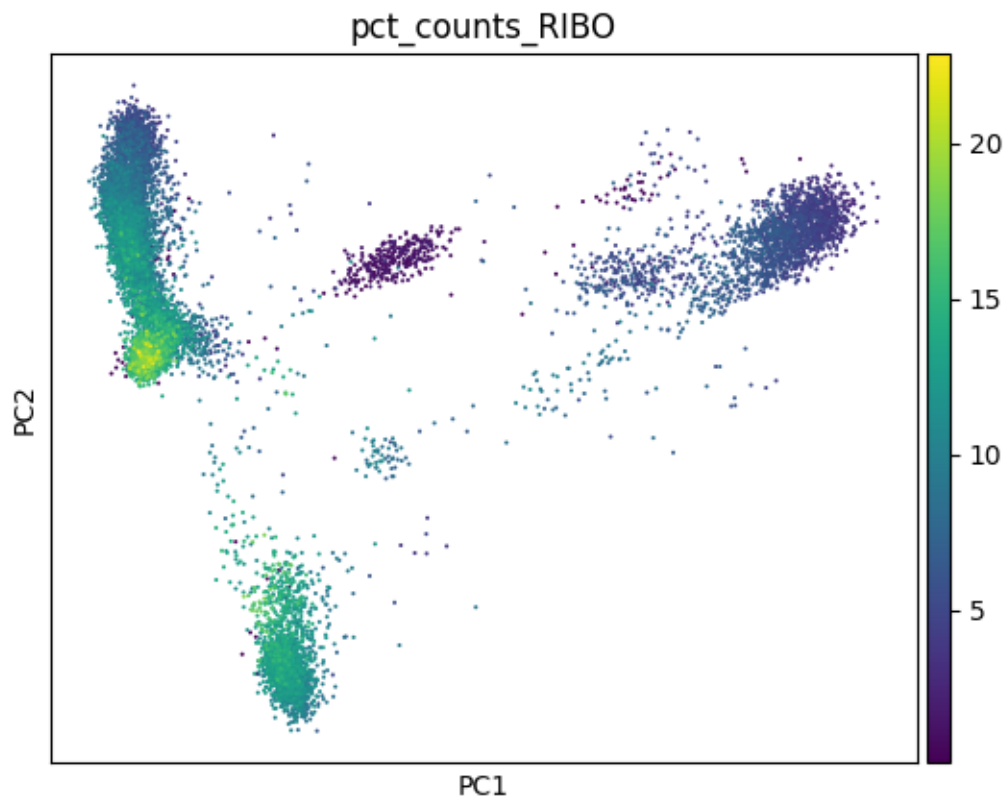
	loglp_total_counts_RIBO	pct_counts_RIBO	total_counts_HB \
AAACCAAAGAACAGAC-1	6.173786	5.075766	0.0
AAACCAAAGCAAGATA-1	7.283448	15.098059	0.0
AAACCAAAGCTGGTTA-1	6.955593	10.238374	0.0
AAACCAGCACCTAACG-1	6.546785	7.147993	0.0
AAACCAGCACGCGTTA-1	7.172425	11.298161	0.0

	loglp_total_counts_HB	pct_counts_HB	n_genes \
AAACCAAAGAACAGAC-1	0.0	0.0	3351
AAACCAAAGCAAGATA-1	0.0	0.0	2718
AAACCAAAGCTGGTTA-1	0.0	0.0	3262
AAACCAGCACCTAACG-1	0.0	0.0	3184
AAACCAGCACGCGTTA-1	0.0	0.0	3342

	doublet_score	predicted_doublet
AAACCAAAGAACAGAC-1	0.021459	False
AAACCAAAGCAAGATA-1	0.017483	False
AAACCAAAGCTGGTTA-1	0.029283	False
AAACCAGCACCTAACG-1	0.011212	False
AAACCAGCACGCGTTA-1	0.095853	False

```
[ ]: sc.pl.pca(
    fresh_blood_adata,
```

```
color=["pct_counts_RIBO"]
)
```



```
[ ]: ## Nearest Neighbour
      # Let us compute the neighborhood graph of cells using the PCA representation
      ↪ of the data matrix.
      # basically we want to cluster the PCA components
```

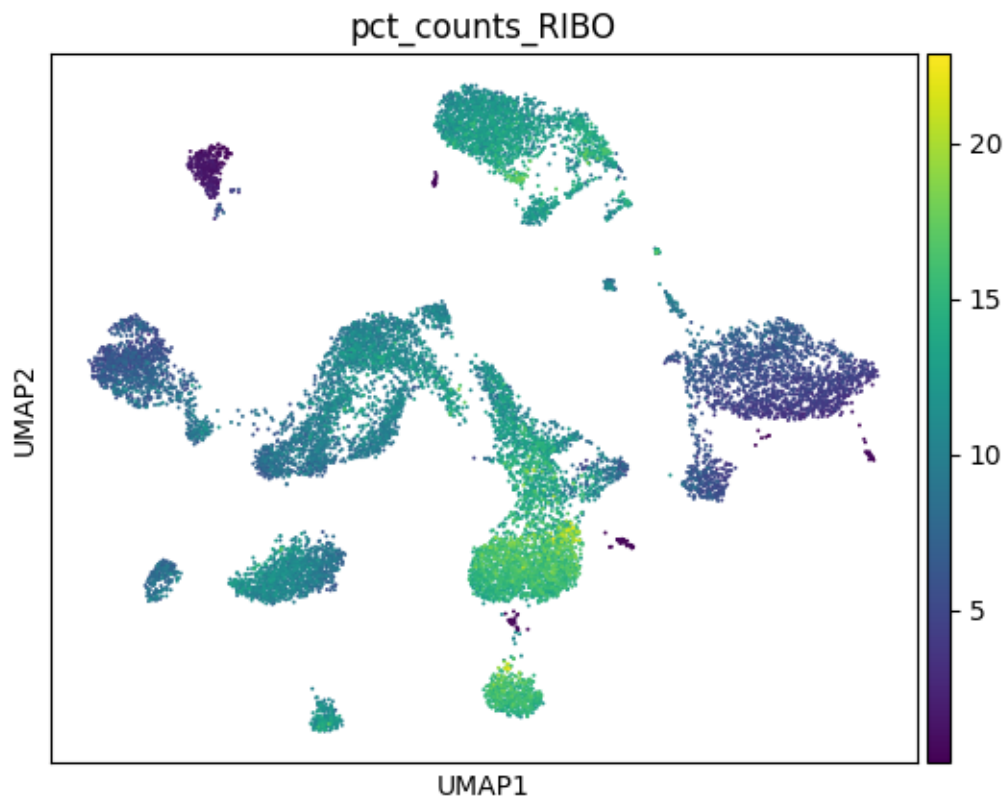
```
[ ]: sc.pp.neighbors(fresh_blood_adata)
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/neighbors/_init_.py:430:
FutureWarning: Use obsm (e.g. `k` in `adata.obsm` or `adata.obsm.keys() | {'u'}`)
instead of AnnData.obsm_keys, AnnData.obsm_keys is deprecated and will be
removed in the future.
    if "X_diffmap" in adata.obsm_keys():
```

```
[ ]: sc.tl.umap(fresh_blood_adata)
```

```
[ ]: sc.pl.umap(
      fresh_blood_adata,
      color=["pct_counts_RIBO"],
```

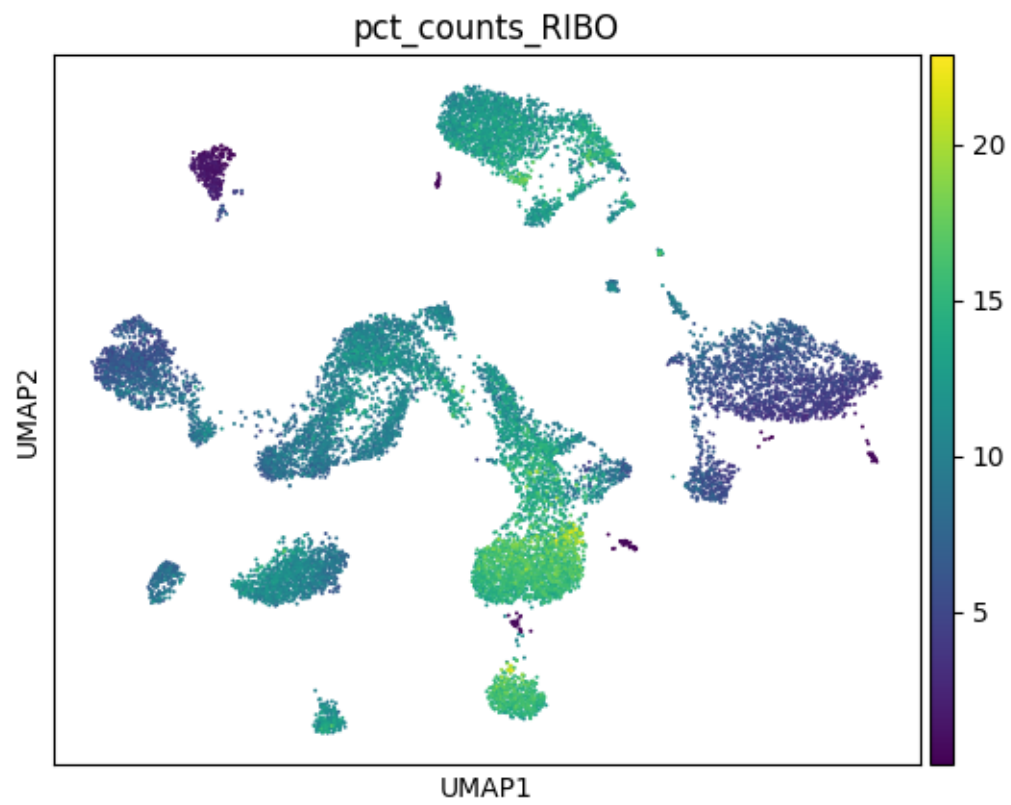
```
size=8,  
)
```



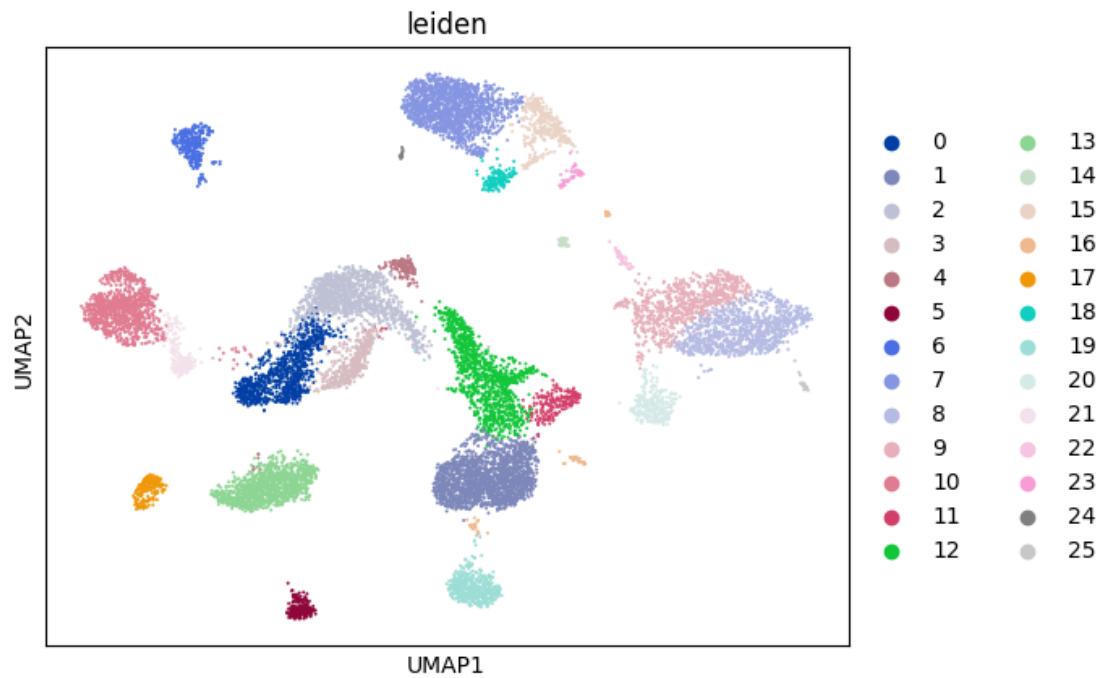
```
[ ]: ## Clustering by communities.  
# Usually used for cell type detection
```

```
[ ]: # Using the igraph implementation and a fixed number of iterations can be u  
# significantly faster, especially for larger datasets  
sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2)
```

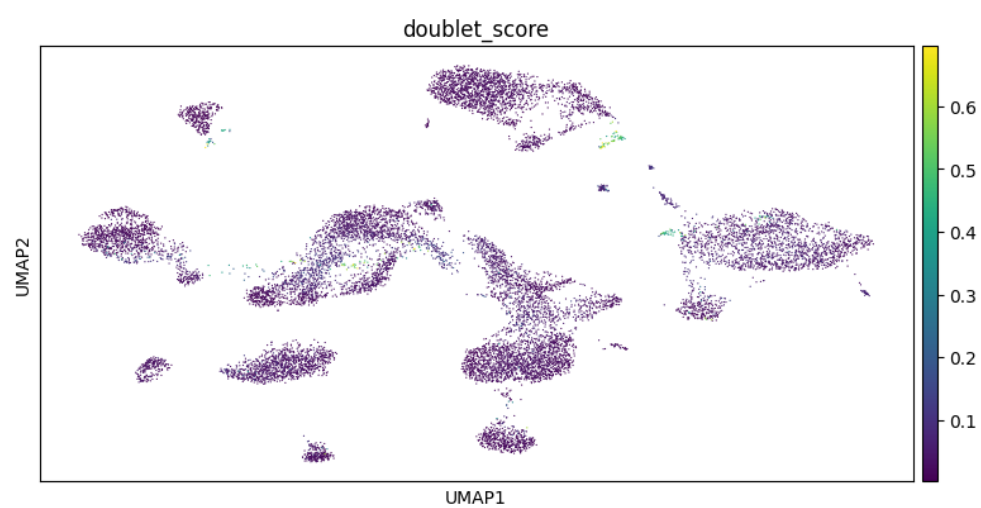
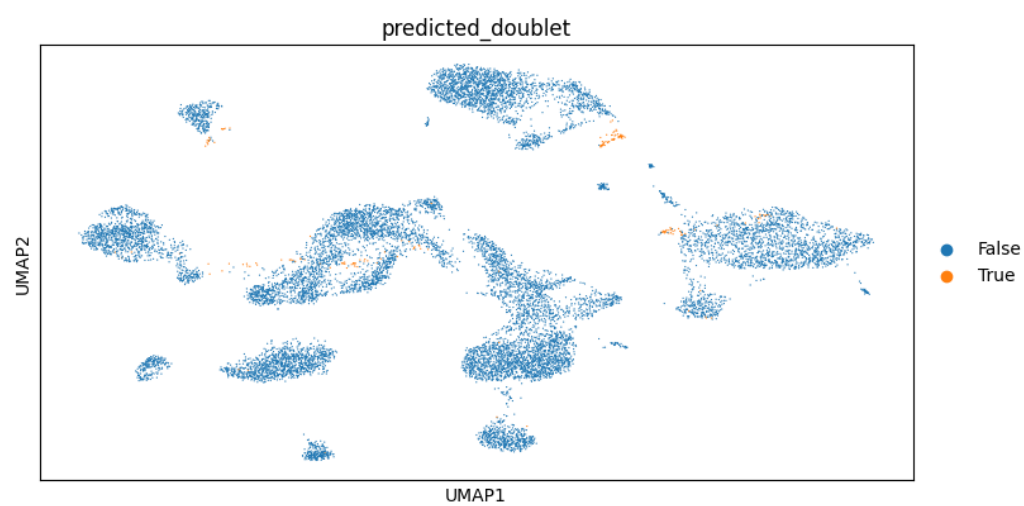
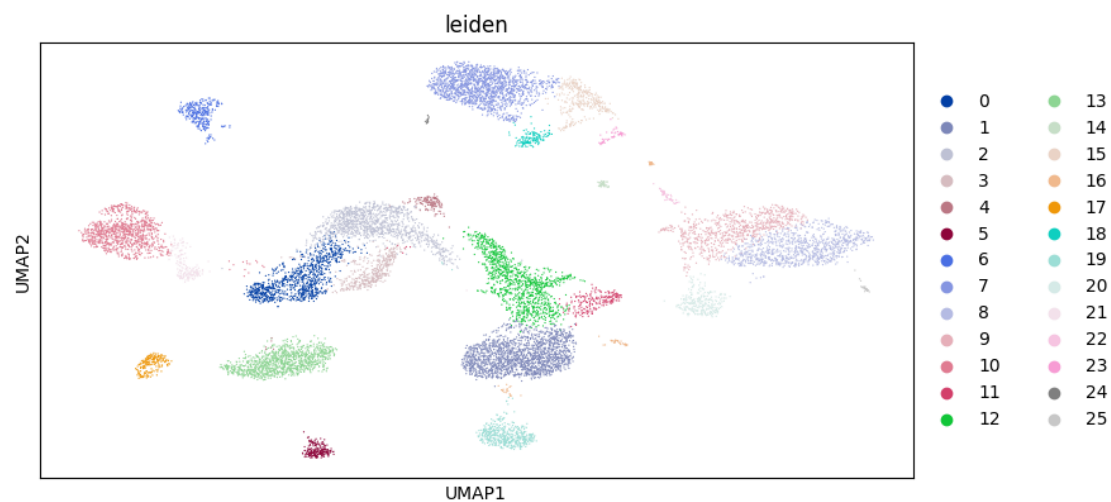
```
[ ]: sc.pl.umap(  
    fresh_blood_adata,  
    color=["pct_counts_RIBO"],  
    size=8,  
)
```

```
[ ]: sc.pl.umap(  
    fresh_blood_adata,  
    color=["leiden"],  
    size=8,  
)
```



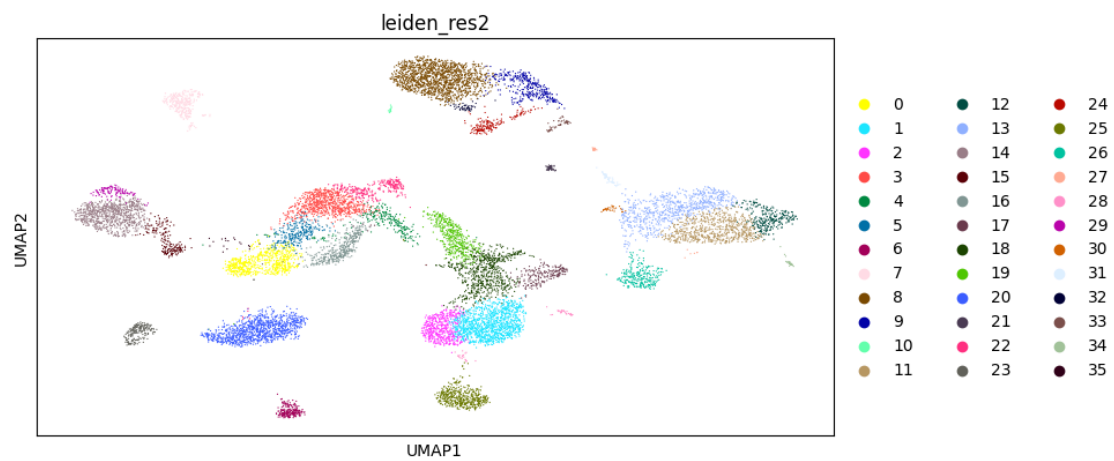
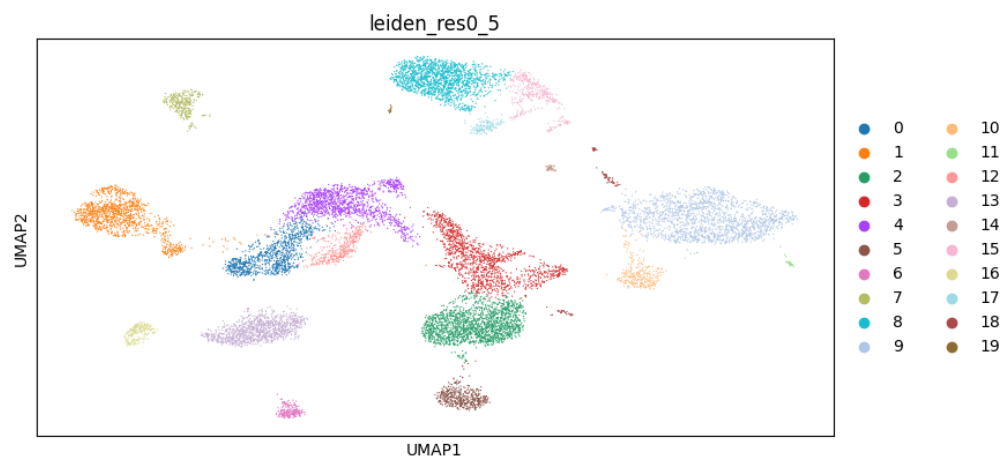
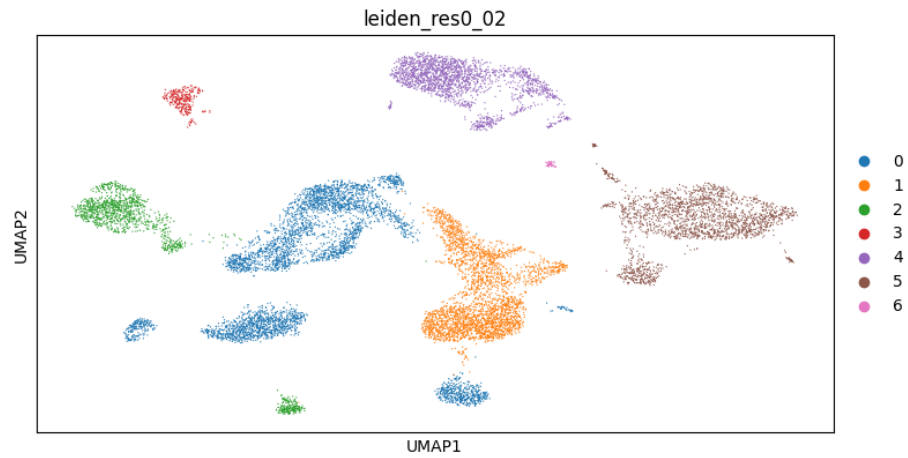
```
[ ]: sc.pl.umap(
    fresh_blood_adata,
    color=["leiden", "predicted_doublet", "doublet_score"],
    # increase horizontal space between panels
    wspace=0.5,
    size=3,
    ncols = 1
)
```



```
[ ]: #Further reclustering
```

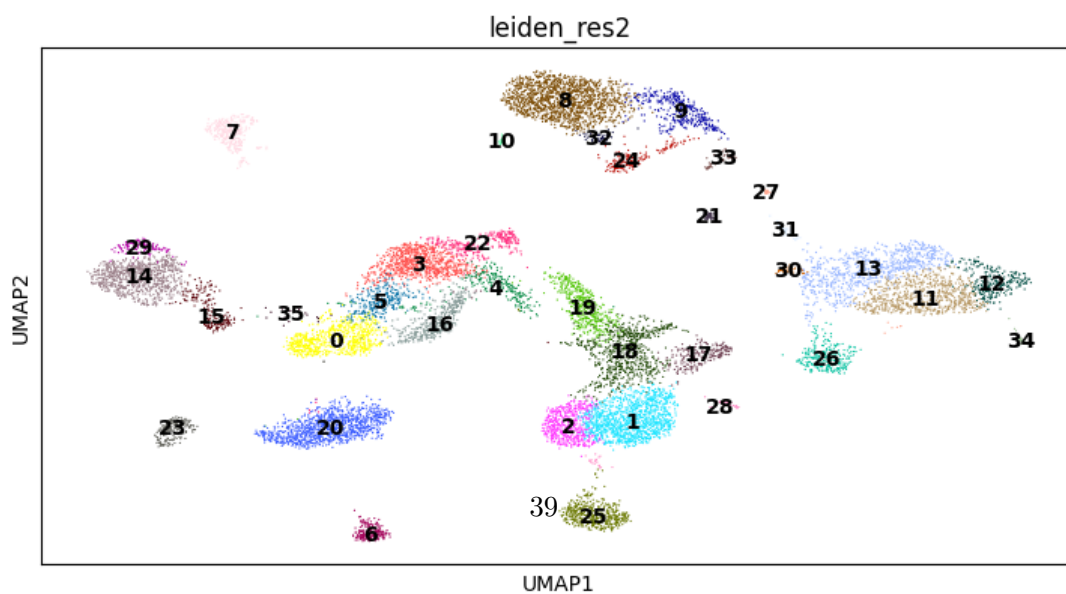
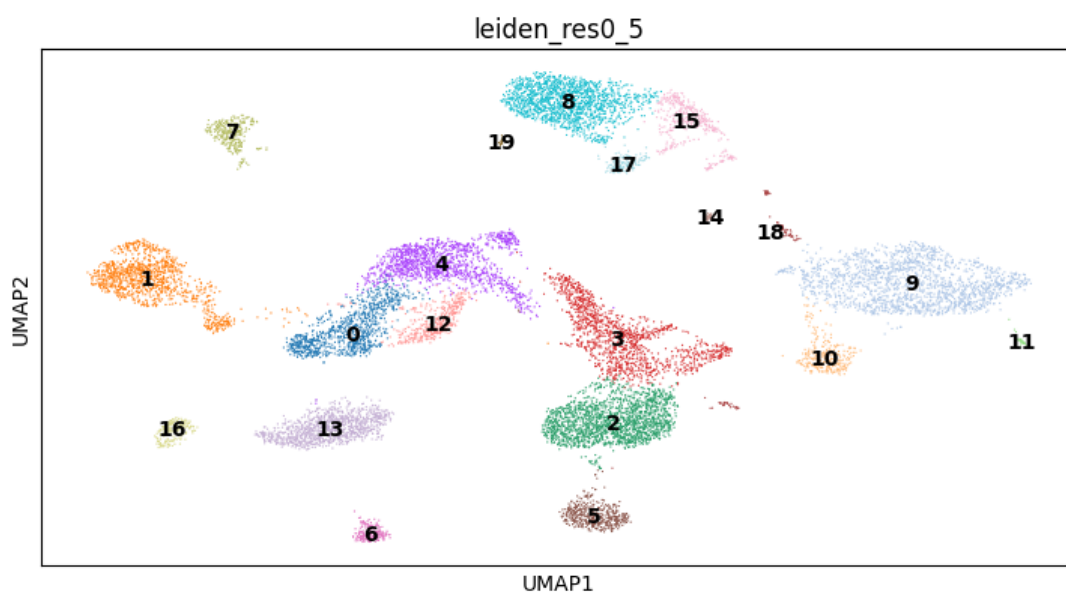
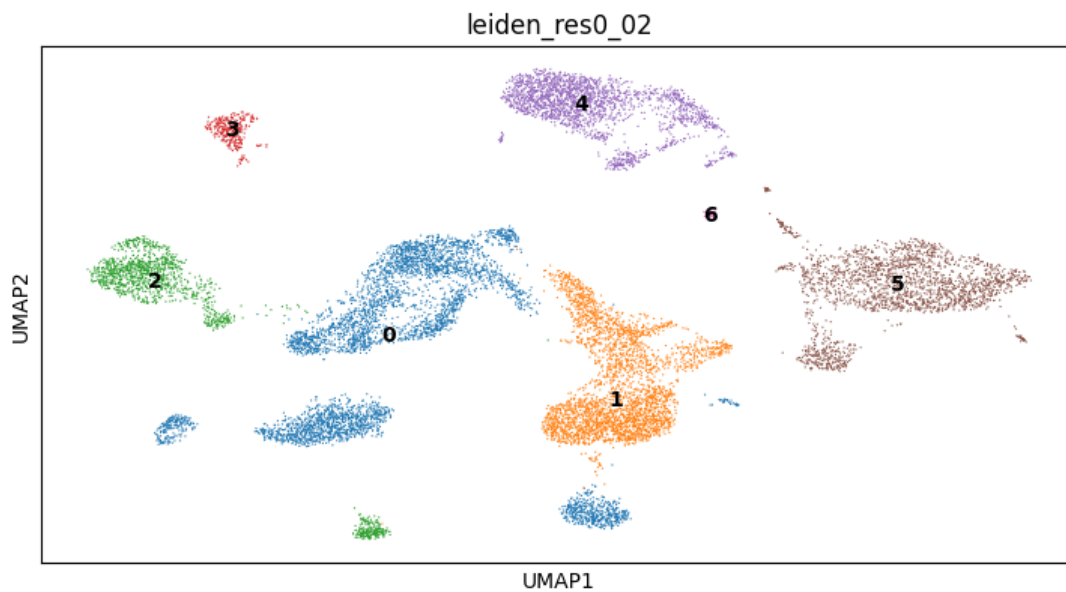
```
[ ]: sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2, ↵  
    ↪key_added="leiden_res0_02", resolution=0.02)  
sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2, ↵  
    ↪key_added="leiden_res0_5", resolution=0.5)  
sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2, ↵  
    ↪key_added="leiden_res2", resolution=2)
```

```
[ ]: sc.pl.umap(  
    fresh_blood_adata,  
    color=["leiden_res0_02", "leiden_res0_5", "leiden_res2"],  
    # increase horizontal space between panels  
    wspace=0.5,  
    size=3,  
    ncols = 1  
)
```



```
[ ]: sc.pl.umap(
    fresh_blood_adata,
    color=["leiden_res0_02", "leiden_res0_5", "leiden_res2"],
```

```
# increase horizontal space between panels  
wspace=0.5,  
size=3,  
ncols = 1,  
legend_loc="on data"  
)
```



```
[ ]: ### Automated cell annotation
```

```
[ ]: import decoupler as dc
```

```
[ ]: # Query Omnipath and get PanglaoDB
markers = dc.op.resource(name="PanglaoDB", organism="human")

# Keep canonical cell type markers alone
markers = markers[markers["canonical_marker"]]

# Remove duplicated entries
markers = markers[~markers.duplicated(["cell_type", "genesymbol"])]

# Format because dc only accepts cell_type and genesymbol

markers = markers.rename(columns={"cell_type": "source", "genesymbol": "target"})
markers = markers[["source", "target"]]

markers.head()
```

```
[ ]:          source  target

2      Bergmann glia    A2M
4          Mast cells  A4GALT
5  Embryonic stem cells  A7KBS4
8      Pinealocytes   AANAT
9      Astrocytes     AANAT
```

```
[ ]: dc.mt.ulm(data=fresh_blood_adata,
              net=markers,
              tmin = 3)
```

```
[ ]: score = dc.pp.get_obs(mf, key="score_ulm")
score
```

```
[ ]: AnnData object with n_obs × n_vars = 13439 × 97
      obs: 'n_genes_by_counts', 'log1p_n_genes_by_counts', 'total_counts',
      'log1p_total_counts', 'pct_counts_in_top_50_genes',
      'pct_counts_in_top_100_genes', 'pct_counts_in_top_200_genes',
      'pct_counts_in_top_500_genes', 'total_counts_MT', 'log1p_total_counts_MT',
      'pct_counts_MT', 'total_counts_RIBO', 'log1p_total_counts_RIBO',
      'pct_counts_RIBO', 'total_counts_HB', 'log1p_total_counts_HB', 'pct_counts_HB',
```



```

'n_genes', 'doublet_score', 'predicted_doublet', 'leiden', 'leiden_res0_02',
'leiden_res0_5', 'leiden_res2'
    uns: 'scrublet', 'log1p', 'hvg', 'pca', 'neighbors', 'umap', 'leiden',
'leiden_colors', 'predicted_doublet_colors', 'leiden_res0_02', 'leiden_res0_5',
'leiden_res2', 'leiden_res0_02_colors', 'leiden_res0_5_colors',
'leiden_res2_colors'
    obsm: 'X_pca', 'X_umap', 'score_ulm', 'padj_ulm'

```

```
[ ]: fresh_blood_adata.obsm["score_ulm"].head()
```

```

[ ]:
      Acinar cells  Adipocytes  Alpha cells  \
AAACCAAAGAACAGAC-1  -1.024520  -1.157857  -1.122553
AAACCAAAGCAAGATA-1  -0.851239  -1.066545   1.392907
AAACCAAAGCTGGTTA-1  -0.972659  -0.605491   1.733040
AAACCAGCACCTAACG-1  -0.975911  -2.175951   1.169892
AAACCAGCACGCGTTA-1  -0.365460  -1.407562   1.452857

      Alveolar macrophages  Astrocytes  B cells  \
AAACCAAAGAACAGAC-1      3.167727   1.788504  0.051880
AAACCAAAGCAAGATA-1     -0.160084   1.674194  1.649956
AAACCAAAGCTGGTTA-1      1.533731   1.245965  1.393795
AAACCAGCACCTAACG-1      1.881583   2.208534 -0.181091
AAACCAGCACGCGTTA-1      1.975010   1.530813  1.212177

      B cells memory  B cells naive  Basophils  Bergmann glia  \
AAACCAAAGAACAGAC-1  -1.991232   -1.919460   0.996606   2.561394
AAACCAAAGCAAGATA-1  -0.657855    0.215860   0.280422  -0.392360
AAACCAAAGCTGGTTA-1  -1.627565   -0.710262  -0.071197   0.367371
AAACCAGCACCTAACG-1  -2.086519   -1.618377   1.766395   2.600667
AAACCAGCACGCGTTA-1  -0.524691   -0.725068  -0.173370   1.918253

      ...  Sebocytes  Sertoli cells  Smooth muscle cells  \
AAACCAAAGAACAGAC-1  ... -0.970895   -0.943990           0.159336
AAACCAAAGCAAGATA-1  ... -1.204066   -1.446825          -1.446825
AAACCAAAGCTGGTTA-1  ... -0.641492    0.093290          -1.218025
AAACCAGCACCTAACG-1  ...  0.863481   -2.325356          -0.313729
AAACCAGCACGCGTTA-1  ...  0.613889   -1.642448          -1.796702

      T cells  T follicular helper cells  T helper cells  \
AAACCAAAGAACAGAC-1  11.967801           -0.348311   3.556622
AAACCAAAGCAAGATA-1   6.544548           -0.851239   1.153128
AAACCAAAGCTGGTTA-1  11.277543           -0.972659   1.850920
AAACCAGCACCTAACG-1   9.256290           -0.975911   3.394022
AAACCAGCACGCGTTA-1  11.714664           -0.365460   2.828105

      T regulatory cells  Tanycytes  Thymocytes  Tuft cells
AAACCAAAGAACAGAC-1     -0.233372  -1.024520   3.257220   2.539718

```

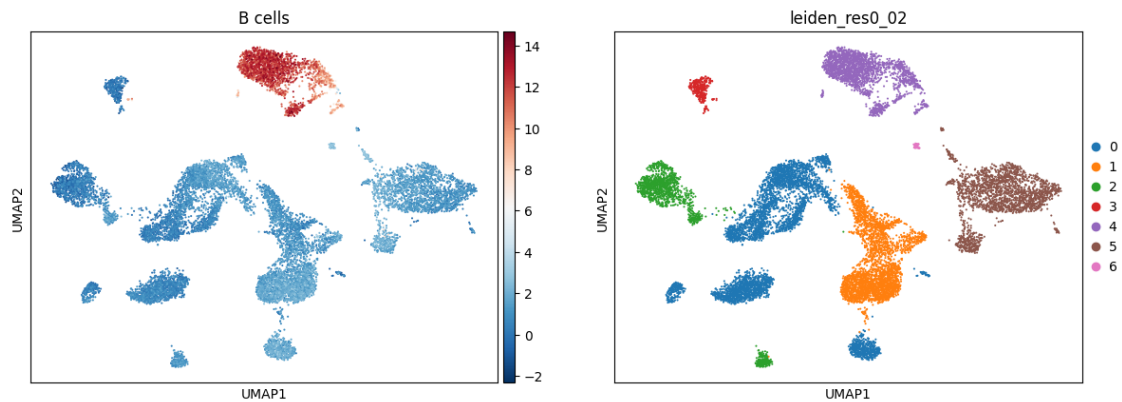
AAACCAAAGCAAGATA-1	-0.715220	-0.160084	1.384593	1.146025
AAACCAAAGCTGGTTA-1	-1.375830	0.065565	1.963522	2.093752
AAACCAGCACCTAACG-1	-1.380429	-0.975911	2.632058	3.212239
AAACCAGCACGCGTTA-1	-1.366579	-0.365460	3.798206	1.923412

[5 rows x 97 columns]

```
[ ]: fresh_blood_adata.obsm["score_ulm"].columns
```

```
[ ]: Index(['Acinar cells', 'Adipocytes', 'Alpha cells', 'Alveolar macrophages',
'Astrocytes', 'B cells', 'B cells memory', 'B cells naive', 'Basophils',
'Bergmann glia', 'Beta cells', 'Cardiomyocytes', 'Cholangiocytes',
'Chondrocytes', 'Chromaffin cells', 'Clara cells', 'Crypt cells',
'Delta cells', 'Dendritic cells', 'Ductal cells',
'Embryonic stem cells', 'Endothelial cells', 'Enterocytes',
'Enteroendocrine cells', 'Eosinophils', 'Ependymal cells',
'Epsilon cells', 'Erythroblasts',
'Erythroid-like and erythroid precursor cells', 'Fibroblasts',
'Gamma (PP) cells', 'Gamma delta T cells', 'Gastric chief cells',
'Germ cells', 'Glomus cells', 'Hematopoietic stem cells', 'Hepatocytes',
'Immature neurons', 'Intercalated cells', 'Keratinocytes',
'Kupffer cells', 'Langerhans cells', 'Leydig cells',
'Loop of Henle cells', 'Luminal epithelial cells', 'Macrophages',
'Mammary epithelial cells', 'Mast cells', 'Megakaryocytes',
'Melanocytes', 'Merkel cells', 'Mesangial cells', 'Microfold cells',
'Microglia', 'Monocytes', 'Myoblasts', 'Müller cells', 'NK cells',
'Natural killer T cells', 'Neural stem/precursor cells', 'Neuroblasts',
'Neurons', 'Neutrophils', 'Nuocytes',
'Oligodendrocyte progenitor cells', 'Oligodendrocytes', 'Osteoblasts',
'Osteoclasts', 'Pancreatic progenitor cells', 'Paneth cells',
'Pericytes', 'Pinealocytes', 'Plasma cells',
'Plasmacytoid dendritic cells', 'Platelets', 'Pluripotent stem cells',
'Podocytes', 'Principal cells', 'Proximal tubule cells',
'Pulmonary alveolar type I cells', 'Pulmonary alveolar type II cells',
'Pyramidal cells', 'Red pulp macrophages', 'Reticulocytes',
'Satellite cells', 'Satellite glial cells', 'Schwann cells',
'Sebocytes', 'Sertoli cells', 'Smooth muscle cells', 'T cells',
'T follicular helper cells', 'T helper cells', 'T regulatory cells',
'Tanocytes', 'Thymocytes', 'Tuft cells'],
dtype='object')
```

```
[ ]: sc.pl.umap(score, color=["B cells", "leiden_res0_02"], cmap="RdBu_r")
```



```
[ ]: import seaborn as sns
```

```
[ ]: #heatmap of all cell types
```

```
sns.clustermap(fresh_blood_adata.obsm["score_ulm"])
```

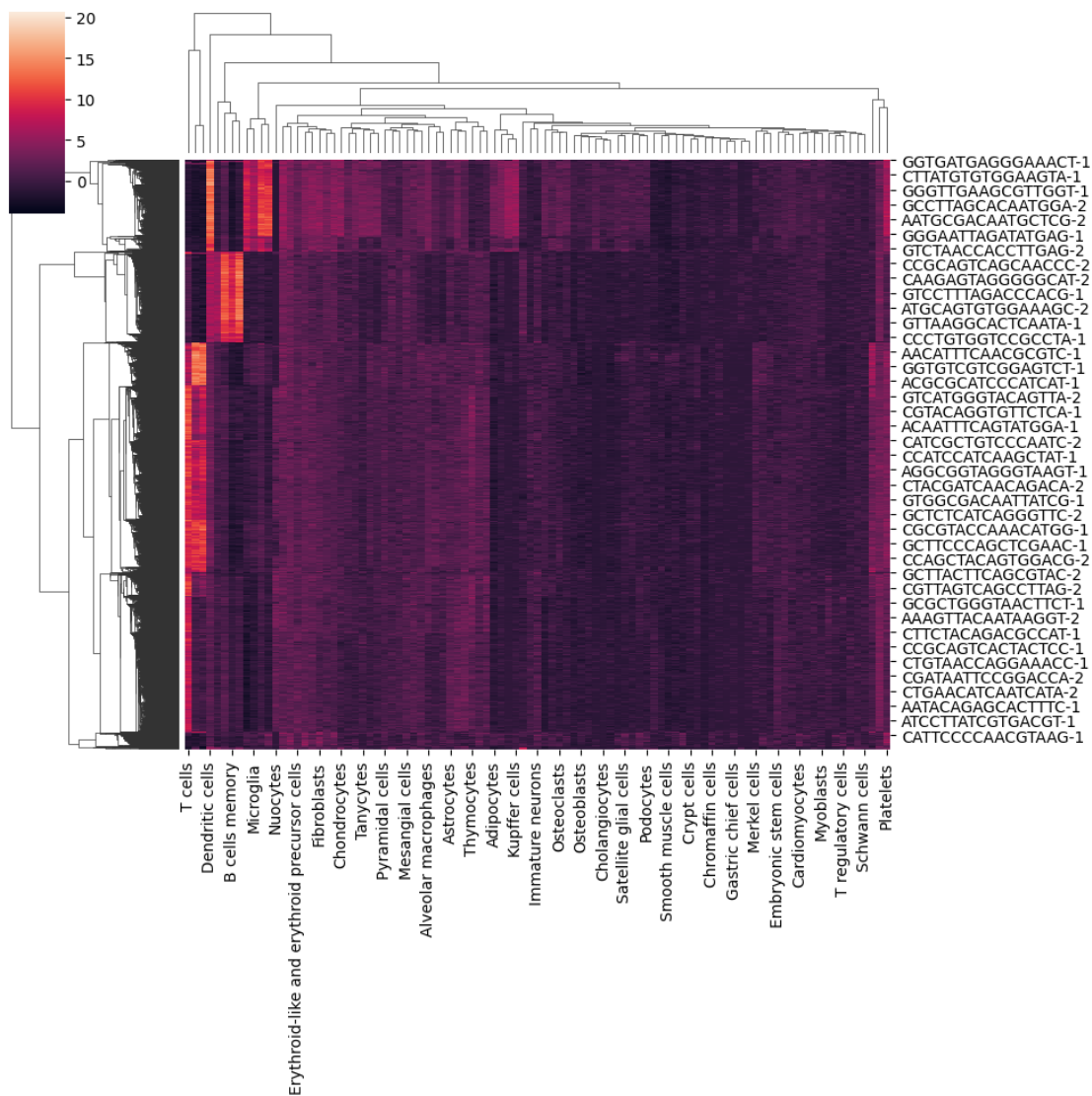
```
/usr/local/lib/python3.12/dist-packages/seaborn/matrix.py:560: UserWarning:
Clustering large matrix with scipy. Installing `fastcluster` may give better
performance.
```

```
warnings.warn(msg)
```

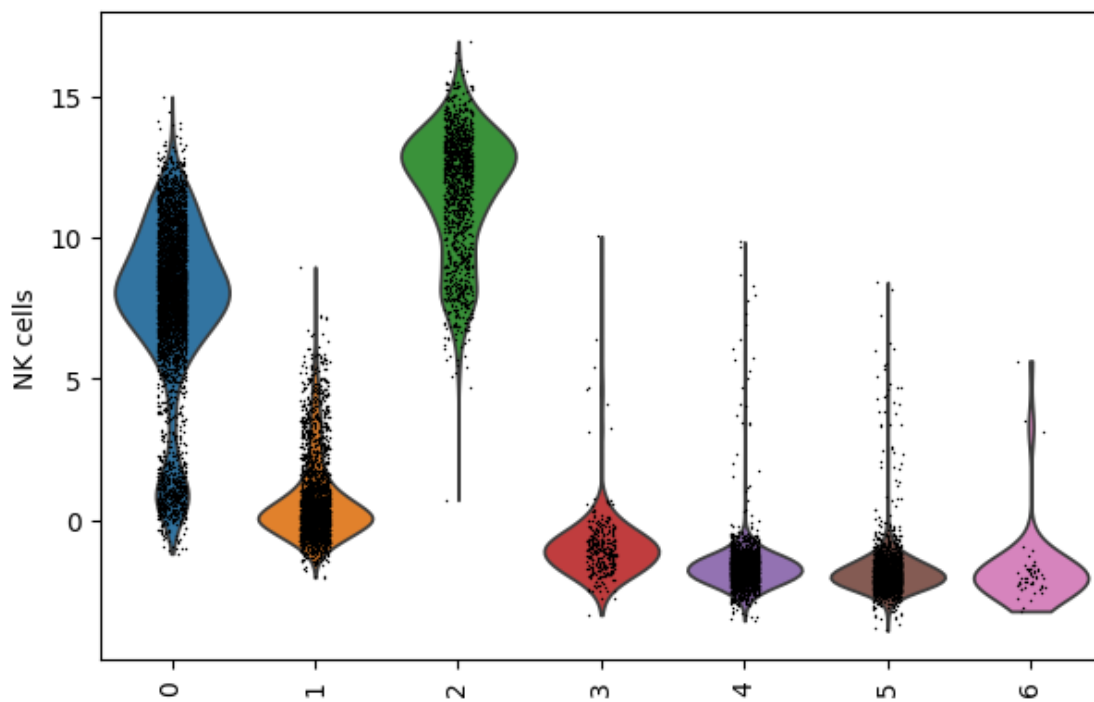
```
/usr/local/lib/python3.12/dist-packages/seaborn/matrix.py:560: UserWarning:
Clustering large matrix with scipy. Installing `fastcluster` may give better
performance.
```

```
warnings.warn(msg)
```

```
[ ]: <seaborn.matrix.ClusterGrid at 0x796bef456b40>
```



```
[ ]: sc.pl.violin(score, keys=["NK cells"], groupby="leiden_res0_02", rotation=90)
```



```
[ ]: #Trajectories
```

```
[ ]: # Example of how to subset for multiple genes in the 'source' column
b_cell_markers = markers[markers['source'].isin(['B cells'])].head()
nk_cell_markers = markers[markers['source'].isin(['NK cells'])].head()
t_cells_markers = markers[markers['source'].isin(['T cells'])].head()

display(b_cell_markers)
```

	source	target
628	B cells	BACH2
633	B cells	BANK1
656	B cells	BCL11A
739	B cells	BST1
1023	B cells	CCR7

```
[ ]: t_cells_markers
```

```
[ ]:      source  target

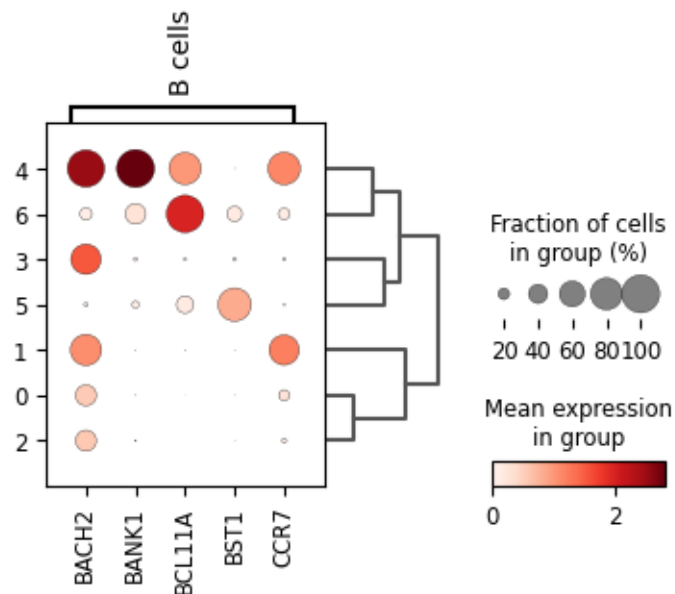
640  T cells   BATF3
733  T cells   BRAF
929  T cells   CCL20
```

```
949 T cells CCL4L1
954 T cells CCL5
```

```
[ ]: marker_genes_dict = {
      "B cells": b_cell_markers['target']
    }
```

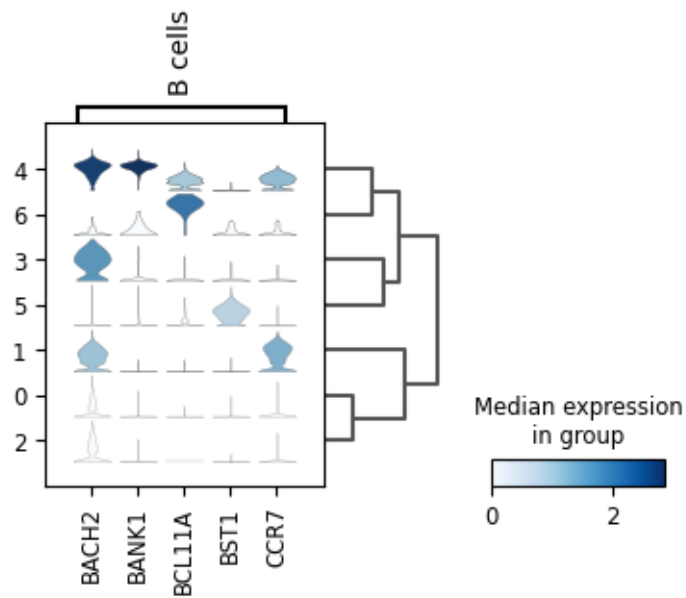
```
[ ]: sc.pl.dotplot(fresh_blood_adata, marker_genes_dict, "leiden_res0_02",
                  dendrogram=True)
```

WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.
categories: 0, 1, 2, etc.
var_group_labels: B cells



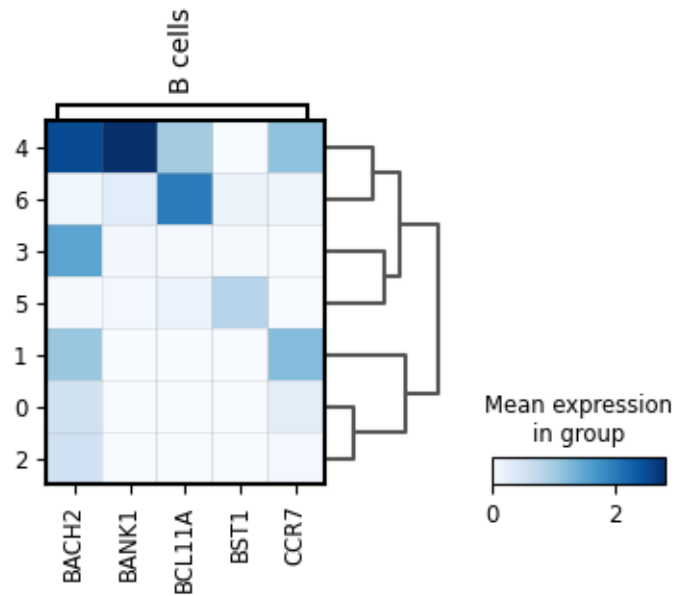
```
[ ]: sc.pl.stacked_violin(
      fresh_blood_adata, marker_genes_dict, groupby="leiden_res0_02",
      dendrogram=True
    )
```

WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.
categories: 0, 1, 2, etc.
var_group_labels: B cells



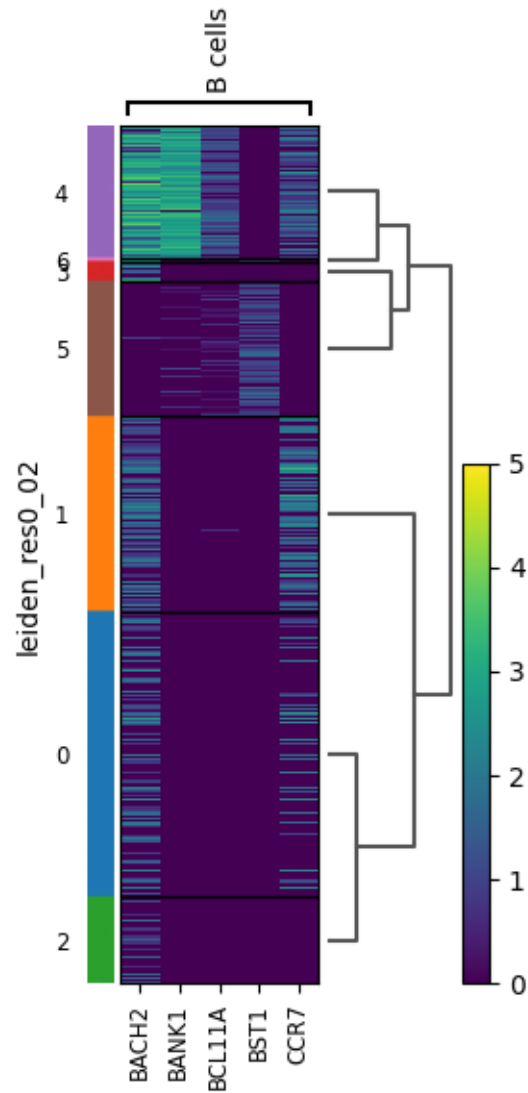
```
[ ]: sc.pl.matrixplot(
    fresh_blood_adata,
    marker_genes_dict,
    "leiden_res0_02",
    dendrogram=True,
    cmap="Blues",
)
```

WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.
categories: 0, 1, 2, etc.
var_group_labels: B cells

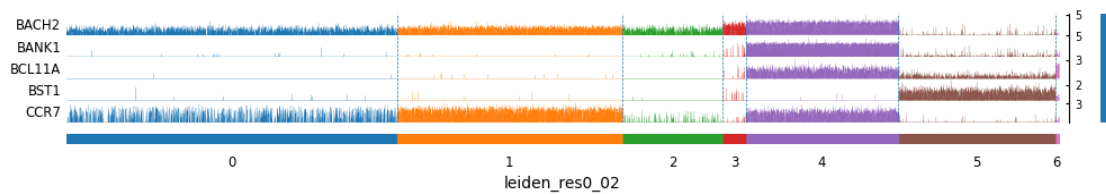


```
[ ]: sc.pl.heatmap(
    fresh_blood_adata, marker_genes_dict, groupby="leiden_res0_02",
    cmap="viridis", dendrogram=True
)
```

WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.
categories: 0, 1, 2, etc.
var_group_labels: B cells



```
[ ]: # @title Default title text
sc.pl.tracksplot(fresh_blood_adata, marker_genes_dict,
↳groupby="leiden_res0_02", dendrogram=False)
```



```
[ ]: #diff exp + Looking for drug targets
```